Adjacency matrices, walks, and connectivity: what $(A^k)_{ij}$ counts, how to test connectedness from A, and what a path between two vertices is

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Abstract

A university lecture whiteboard gives a graph with N vertices by its adjacency matrix A and asks two questions about it; the assignment text adds a third. This paper answers all three for finite simple undirected graphs. It follows the definitions and the numbering of the course textbook (for example Definition 1.2.10 for a walk) [9]. First, the entry $(A^k)_{ij}$ of the k-th power of A is the number of *walks* of length k from vertex n_i to vertex n_j; it does not count paths. The paper proves this by induction on k (the statement is also in the University of Galway lecture notes [2]), and shows on the board matrix why the difference between walks and paths matters. Second, for $N \geq 2$ the graph is connected if and only if every entry off the diagonal of $A + A^2 + \cdots + A^{N-1}$ is positive, equivalently if and only if $(I + A)^{N-1}$ has no zero entry. The paper proves both forms itself, shows that $N - 1$ cannot be lowered to $N - 2$, and states breadth-first search as the alternative. Third, in the course convention a path between n_i and n_j is a walk from n_i to n_j in which no vertex repeats; this is the textbook's Definition 1.2.13, which the textbook calls a "simple path". The textbook's own word "path" (Definition 1.2.12) means something weaker, and this paper calls that object a trail. All matrices in the worked examples were computed by script.

1 Introduction

1.1 The three questions and where each came from

A university lecture whiteboard, photographed twice from slightly different angles, is the starting point of this paper [4, 5]. Read from top to bottom, the board shows six things (Table 1). They set up a graph with N vertices that is given only by its adjacency matrix A, and they ask two questions about it. The words of the problem statement are the board's own: "Given graph (G, V) with N nodes i.e. $|G| = N$ and the graph is represented through it's adj. matrix. So you have A matrix" [4, 5].

The paper answers three questions.

Q1. For a positive integer k, what does the entry $(A^k)_{ij}$ of the k-th power of A represent? This is the board's first question [4, 5].

Q2. How can one tell from A alone whether the graph is connected? This is the board's second question [4, 5].

Q3. What is the definition of a path between two vertices n_i and n_j? This question is not on the board. It was added by the assignment text that came with the photographs. That text states the course convention: in the course, "path" means a walk with no repeated vertices. This is the sentence the answer to Question 3 rests on (Section 5).

Table 1: Every line of the board, in reading order, and where this paper treats it.

No.	What the board shows	Where treated
1	$A_{ij} = 1$ if $e_{ij} \in V$, and 0 if $e_{ij} \notin V$ (the board's V is the edge set)	Definition of the adjacency matrix, Section 2; letters in Section 1.3
2	A drawing of three circled vertices $1, 2, 3$ and a grid labelled "adj." with rows 010, 101, 010	Mismatch reported in Section 1.3; worked example in Section 6
3	"Problem. Given graph (G, V) with N nodes ... So you have A matrix"	Setting, Section 1.3
4	Question 1: "What does A_{ij} represent", with a small mark above A (a power k; see item 2 of Section 1.3)	Section 3
5	Question 2: "How to know if graph is connected or not"	Section 4
6	"Def. Graph is called connected if for any n_i, n_j $\exists$ path between nodes"	Definition of connected, Section 2; the word "path" in Section 5

Question 3 matters because the board's definition of "connected" uses the word "path" without saying which of several meanings it has (Section 1.3), so the answer to Question 2 could appear to depend on it. It does not (Proposition 2.59), but Question 3 still needs a definite answer.

Statements in this paper that are not plain facts from a source carry a label in square brackets. *[derived here]* marks a result proved or computed in this paper; *[interpretation]* marks a reading of the material; *[estimate]* marks an operation count; *[qualified]* marks a statement that rests on something the sources do not fully settle.

The course textbook, *Graph Theory* by Petrosyan, Mkrtchyan and Kamalyan [9], supplies the definitions and their numbers. It contains neither the walk-counting theorem nor a matrix test for connectivity in the pages that were checked *[qualified]* (Section 1.6 says which pages). So the paper proves both results itself and marks them *[derived here]*.

The answers in brief. The details and the proofs follow; this list is only a map.

A1. $(A^k)_{ij}$ is the number of *walks* of length k from n_i to n_j (Theorem 3.5). It is not the number of paths: walks may repeat vertices and edges (Proposition 3.10).

A2. For $N \geq 2$, the graph is connected if and only if every entry off the diagonal of $A + A^2 + \cdots + A^{N-1}$ is positive (Theorem 4.6). Equivalently, $(I + A)^{N-1}$ has no zero entry (Proposition 4.16). The exponent $N - 1$ cannot be lowered to $N - 2$ (Proposition 4.17).

A3. In the course convention, a path between n_i and n_j is a walk from n_i to n_j whose vertices are pairwise distinct (Definition 5.1). It is the textbook's "simple path", Definition 1.2.13 [9].

1.2 The story: houses and roads

In plain words. Picture a small village. Each house is a point. Each road joins exactly two different houses. Every road can be driven in both directions. No road starts and ends at the same house. No two roads join the same pair of houses. The village has only finitely many houses. A trip is a list of houses, one after another. Each house on the list shares a road with the house before it. A trip may pass the same house twice. It may drive the same road twice. A trip may also be just one house, with no driving at all. A trail is a trip that never uses the same road twice. A route is a trip that never passes the same house twice. Every idea in this paper is about this village and its trips.

The village from the board has three houses in a row. The first house and the middle house share a road. The middle house and the last house share a road. No road joins the first house and the last house. Figure 1 (left) shows this village. The same story is used before every definition, theorem, test and example, so that each can be followed without the symbols. Table 2 gives the dictionary from the story to the textbook's words and numbers. The story words are fixed: *trip* for a walk, *trail* for a trail, *route* for a path.

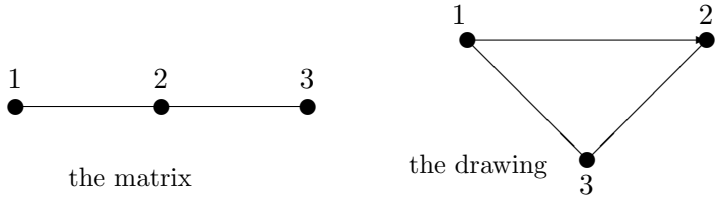


Figure 1: Left: the village of the board matrix, three houses in a row, with roads 1–2 and 2–3. Right: the board drawing as it appears in both photographs, a line from 1 to 2 ending in an arrowhead at 2, a line from 1 to 3, and a line from 2 to 3 [4, 5]. The two pictures do not describe the same graph; see Section 1.3.

Table 2: The story and the technical words. Textbook numbers are from the course textbook [9]; "n.n." means that the pages checked give no number.

Story word	Technical term	Textbook number	Defined in
house	vertex	1.1.1	Definition 2.1
road	edge	1.1.1	Definition 2.1
two houses share a road	adjacent	1.1.4	Definition 2.5
table of ones and zeros	adjacency matrix	n.n. (p. 11)	Definition 2.9
number of roads at a house	degree	1.2.1	Definition 2.16
trip	walk	1.2.10	Definition 2.23
trip that ends where it began	closed walk	1.2.11	Definition 2.27
trip that never uses a road twice	trail	1.2.12	Definition 2.31
route: a trip that never passes a house twice	path	1.2.13	Definition 2.35
round trip that never uses a road twice	cycle	1.2.14	Definition 2.40
round trip that passes no house twice but the first	simple cycle	1.2.15	Definition 2.44
fewest roads on a trip between two houses	distance	1.2.16	Definition 2.48
village in one piece	connected graph	2.1.1	Definition 2.55
one piece of a village	component	n.n. (p. 49)	Definition 2.60

1.3 Conventions and assumptions

Every reading below is a choice made where the photographs, the assignment text, or the textbook leave room. Each is stated here so that the reader can see what the rest of the paper rests on.

1. **Setting.** The graph is finite, simple (no loops, no repeated edges) and undirected. This is the assumption of the assignment text. It agrees with the board matrix, which is symmetric and has zeros on the diagonal (item 4 below), and with the textbook's definition of a graph [9, Section 1.1, Definition 1.1.1, p. 9].

2. **The mark above A in Question 1.** In both photographs the mark above A is small. It looks like quotation marks or a small n, and it is not clearly a k [4, 5]. This paper reads it

as a power k, because the assignment text asks about A^k. This reading is *[interpretation, qualified]*. The mathematics does not depend on the letter: the theorem in Section 3 holds for every positive integer exponent, whether the board meant k or n.

3. **Quotation marks around A.** Two lines lower, the board writes "A" ("So you have "A" matrix") [4, 5]. This paper takes it as the name of the adjacency matrix.

4. **The drawing and the matrix disagree.** The drawing shows three lines: 1–2 with an arrowhead at 2, 1–3, and 2–3 [4, 5]. The matrix next to it has $A_{12} = A_{21} = 1$ and $A_{23} = A_{32} = 1$, and $A_{13} = A_{31} = 0$ (Table 3). So the drawing has an edge 1–3 that the matrix does not have, and an arrowhead (a direction) that the symmetric matrix does not have. The problem statement says the graph *is given by its adjacency matrix* [4, 5], so this paper takes the matrix as the example graph and calls it B (Figure 1, left). It treats the arrowhead as a stray mark, since the matrix is symmetric and the problem is about undirected graphs. The paper reports the mismatch and does not guess which one the lecturer intended. The three-line drawing is called T (a triangle) and is used only where a graph with a cycle is needed.

5. **The board's letters.** The board writes (G, V) for the graph and $e_{ij} \in V$ in the definition of A_{ij}, so the board's V is the *edge* set. The textbook writes $G = (V, E)$, with V for vertices and E for edges [9]. This is a difference of notation only. The paper uses the textbook's V and E. The board writes $|G| = N$ for the number of vertices and n_i for vertices; the paper keeps N and n_i. The textbook writes v_i and n; these are mentioned only here.

6. **The word "path".** The textbook uses "path" for a walk with no repeated edge (Definition 1.2.12) and "simple path" for a walk with no repeated vertex (Definition 1.2.13) [9]. English texts call the first a trail and the second a path [3, 8]. The assignment text says that in the course "path" means no repeated vertices. The board does not say which meaning it uses. This paper therefore writes *trail* for Definition 1.2.12 and *path* for Definition 1.2.13 everywhere, and shows the three conventions side by side in Table 7. That the course uses this meaning is *[interpretation, qualified]*, because it comes from the assignment text and not from the board.

7. **Lengths and exponents.** The exponent k is a positive integer. The case $k = 0$ (the identity matrix, one walk of length 0 from a vertex to itself) is treated by Grinberg [3] and is not needed for the board question. A walk of length 0, a single vertex, is allowed, because the textbook's own property $d(u, u) = 0$ needs it. The textbook does not say so in words, so this is *[interpretation, qualified]*.

Table 3: The board drawing and the board matrix, edge by edge [4, 5].

Pair of vertices	In the drawing	In the matrix	Verdict
$1, 2$	line, arrowhead at 2	$A_{12} = A_{21} = 1$	edge in both; arrowhead has no counterpart
$2, 3$	line	$A_{23} = A_{32} = 1$	edge in both
$1, 3$	line	$A_{13} = A_{31} = 0$	edge in the drawing only

1.4 Notation

The letters are fixed here, before the first definition, and used unchanged after this point. Table 4 also gives the board's and the textbook's letters.

Table 4: Notation of this paper, with the board's and the textbook's letters.

Symbol	Meaning	Board	Textbook
$G = (V, E)$	a finite simple undirected graph, vertex set V, edge set E	(G, V), with V the edge set	$G = (V, E)$
N	number of vertices	$\lvert G\rvert = N$	n
$n_1, \ldots, n_N$	the vertices	n_i	v_i
u, v, w	any vertices	—	u, v
$n_i n_j$, e_{ij}	the edge joining n_i and n_j	e_{ij}	uv
A, A_{ij}	adjacency matrix and its entry in row i, column j	A, A_{ij}	$A(G)$, a_{ij}
A^k, I	k-th power of A and the $N \times N$ identity matrix	A with a small mark above	—
k, m	positive integers: lengths of walks, exponents	the mark above A	—
i, j, r	indices running from 1 to N	i, j	—
$\deg(n_i)$	degree of n_i (Definition 1.2.1)	—	$d_G(v)$, $d(v)$
$d(u, v)$	distance	—	$d(u, v)$, $d_G(u, v)$
S	$A + A^2 + \cdots + A^{N-1}$, defined in Section 4	—	—

1.5 How the paper is organized

Section 2 gives the definitions in the textbook's order. Each definition is followed by a remark on what it allows and forbids, an example and a non-example. Sections 3 to 5 answer Questions 1, 2 and 3 in that order. Section 6 writes out every matrix for the board graph B and for a graph with a single edge. Section 7 states limitations and Section 8 answers the three questions directly. Appendix A holds derivations and the scripts, and Appendix B lists the course terms with their textbook numbers.

Every definition, theorem, lemma, test, and worked example is preceded by a short paragraph marked "In plain words". A source that is cited states the fact itself; where only the statement is cited and the proof is the paper's own, the text says so.

1.6 Provenance: sources, pages checked, and scripts

This subsection collects where the material comes from, so that Section 2 to Section 6 can stay mathematical.

Sources. The board photographs [4, 5]; the course textbook [9]; Grinberg's lecture-note text on graph theory [3]; the University of Galway lecture notes on the adjacency matrix [2]; the open textbook of Lehman, Leighton and Meyer [6]; the journal paper of Paramadevan and Sotheeswaran [7]; Bai's lecture notes [1]; the MathWorld and Wikipedia entries [11, 8]; and lecture slides on breadth-first search [10]. The Galway file carries no author name or date; its URL suggests the name Quinlan, which the file does not confirm, so this paper calls it the University of Galway notes.

Pages of the textbook checked. The textbook definitions were taken from printed pages 9 to 12 (graphs and the adjacency matrix), 16 (degree), 21 to 23 (walks, trails, paths, cycles, distance, Lemma 1.2.1) and 48 to 49 (connectedness). The remaining pages of Chapters 1 and 2 (13 to 15, 17 to 20, 24 to 47 and 50 to 62) and Chapters 3 to 8 were not read page by page. An independent full-text search of the whole book for the adjacency matrix found it only on pages

11 to 12, 34 and 60, and found no walk-counting theorem and no matrix connectivity test. So the statement "the textbook contains neither" is *[qualified]*: it rests on that search and on the pages read, not on a page-by-page reading of the whole book. Page numbers in this paper are printed page numbers; the PDF page is one higher.

Weaknesses of some sources. Two sources that state the connectivity test prove it loosely. Paramadevan and Sotheeswaran's Theorem 1 reads: "the number of different paths of length l between the i-th and j-th vertices of G is equal to the (i, j)-th entry of A_G^l". This is false for paths in either sense (Proposition 3.10), although their own property 7 says walks. Their proof of Theorem 2 repeats the word: "b_{ij} = number of different paths of length less than n from the vertex i to j", and it ends "Conversely, if $b_{ij} \neq 0$, then there exist at least one path from v_i to v_j, thus G is connected", which passes from a non-zero entry to a path without the walk-to-path lemma [7]. Bai's proof sketch of the $(I + A)^{n-1}$ form says "$(I + A)^{n-1}$ aggregates all paths of length $\leq n - 1$" [1]. Both statements of the test are correct and the proofs are incomplete *[interpretation]*. This is why the paper proves the test itself and cites these two sources for the statements only.

Scripts. Every matrix, product, sum and count in the paper was computed by script before it was written. The scripts, their saved outputs, and the script that turns those outputs into the matrices printed in the text are in `appendices/scripts/`; Appendix A.7 says how to run them. Exhaustive comparisons over small graphs are checks, not proofs, and the text says so each time.

2 Background: the definitions, in the textbook's order

Let $G = (V, E)$ be a finite simple undirected graph with N vertices $n_1, \ldots, n_N$; let u, v, w be vertices and e an edge. These are the letters of Table 4, and every definition below uses exactly them. The definitions follow the order of the course textbook [9]: graph, adjacency, adjacency matrix, degree, walk, closed walk, trail, path, cycle, simple cycle, distance, and then connectedness. Each definition cites the textbook's number and printed page. After each definition come a remark on what it allows and forbids, an example, and a non-example.

Three graphs supply the examples. The first is the graph B of the board matrix: vertices $1, 2, 3$ (meaning n_1, n_2, n_3) and edges 12 and 23 (Figure 1, left). The second is the board drawing T, a triangle with edges 12, 13, 23. The third is the textbook's own example graph, which this paper calls G_{tb} (Figure 2); the seven edges used from it are v_1v_2, v_1v_3, v_2v_3, v_2v_4, v_3v_4, v_3v_5 and v_4v_5, the ones that the textbook's own examples force. It is needed because B has no cycle, and in B every trail is a path (Remark 2.32), so B cannot show the differences between the later definitions.

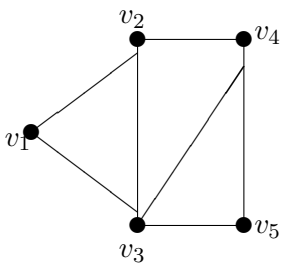


Figure 2: The graph G_{tb} with the seven edges that the textbook's own examples force: v_1v_2, v_1v_3, v_2v_3, v_2v_4, v_3v_4, v_3v_5, v_4v_5. It is a part of the textbook's Fig. 1.2.5 (printed p. 22) [9]. The textbook's figure may have further edges; nothing in this paper depends on them.

2.1 Graphs and the adjacency matrix

> **In plain words.** A village is a list of houses and a list of roads. Each road joins two different houses. Two roads never join the same pair. Roads have no direction. The list of houses is finite and has at least one house.

Definition 2.1 (Simple undirected graph; textbook Definition 1.1.1, p. 9)**.** A *simple undirected graph* is a pair $G = (V, E)$, where V is a finite nonempty set, the *vertices*, and E is a set of unordered pairs of distinct vertices, the *edges*. The edge $\{u, v\}$ is written uv [9, Section 1.1, Definition 1.1.1, p. 9].

Remark 2.2. Because the pair must consist of distinct vertices, there is no loop (an edge from a vertex to itself). Because E is a set, no pair occurs twice, so there is no repeated edge. Because the pair is unordered, uv and vu are the same edge. Isolated vertices (vertices with no edge) are allowed, since E may be any set of such pairs. An edge is fixed by its two ends. This is why a walk can later be written by its vertices alone (Remark 2.24). In this paper the word *loop* always means such a self-edge.

Example 2.3. $V = \{n_1, n_2, n_3\}$ and $E = \{n_1n_2,\, n_2n_3\}$. This is the graph B of the board matrix, three houses in a row.

Non-example 2.4. Two houses joined by two roads, or a road from one house back to itself. The first repeats an edge, the second is a loop. Neither is a simple graph.

> **In plain words.** Two houses are neighbours when one road joins them directly. A house is never its own neighbour. Being neighbours says nothing about longer trips.

Definition 2.5 (Adjacent; textbook Definition 1.1.4, p. 11)**.** Vertices u and v are *adjacent* if $uv \in E$ [9, Definition 1.1.4, p. 11]. Then each is called a *neighbour* of the other.

Remark 2.6. Adjacency is symmetric ($uv = vu$), and no vertex is adjacent to itself, since there are no loops. It says nothing about longer connections.

Example 2.7. In B, the vertices n_1 and n_2 are adjacent.

Non-example 2.8. In B, the vertices n_1 and n_3 are not adjacent, although a trip of two roads joins them.

> **In plain words.** Make a square table with one row and one column for every house. Write a one where the row house and the column house share a road. Write a zero everywhere else. This table describes the whole village.

Definition 2.9 (Adjacency matrix; textbook p. 11, after Definition 1.1.6; board, first line)**.** A *matrix* is a rectangular table of numbers; its entry in row i and column j is written with the indices i, j. The *adjacency matrix* of G is the $N \times N$ matrix A with $A_{ij} = 1$ if n_i and n_j are adjacent, and $A_{ij} = 0$ otherwise [9, p. 11]. The board writes the same rule as $A_{ij} = 1$ if $e_{ij} \in V$ and 0 if $e_{ij} \notin V$, with its V the edge set [4, 5].

Remark 2.10. The definition allows only the entries 0 and 1. It depends on how the vertices are numbered: renumbering permutes rows and columns together. *[derived here]*

Example 2.11. The board matrix, with rows (0 1 0), (1 0 1), (0 1 0):

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

is the adjacency matrix of B, since $A_{12} = A_{21} = 1$, $A_{23} = A_{32} = 1$ and all other entries are 0.

Non-example 2.12. The matrix with rows (0 1 0), (0 0 1), (0 1 0) is not the adjacency matrix of any simple undirected graph: its entry A_{12} is 1 but A_{21} is 0, which would need a one-way road. A matrix with a 1 on the diagonal (a loop) or an entry 2 (a double road) is not one either.

> **In plain words.** The table has some easy features. It looks the same when flipped over its diagonal. The diagonal holds only zeros. And any table of ones and zeros with these two features comes from some village.

Proposition 2.13 (Properties of A)**.** *For the adjacency matrix A of a simple undirected graph:*

1. *every entry is* 0 *or* 1*;*
2. *A is symmetric: $A_{ij} = A_{ji}$ for all i, j;*
3. *the diagonal is zero: $A_{ii} = 0$ for all i;*
4. *conversely, every $N \times N$ matrix of zeros and ones that is symmetric with zero diagonal is the adjacency matrix of some simple graph with numbered vertices.*

These are stated in the textbook [9, pp. 11–12].

Proof. Items 1 to 3 are read off Definition 2.9: adjacency is symmetric and irreflexive (Remark 2.6). Item 4: join n_i and n_j exactly when the entry is 1. Symmetry makes this a legitimate set of unordered pairs, and the zero diagonal excludes loops. □

Example 2.14. The board matrix is symmetric with zero diagonal, as items 2 and 3 require.

Non-example 2.15. The matrix with rows (0 1 0), (1 0 1), (1 1 0) is not symmetric (its entry A_{31} is 1 and A_{13} is 0), so it is not an adjacency matrix.

> **In plain words.** The busy-ness of a house is the number of roads that meet there. We call that number the degree of the house.

Definition 2.16 (Degree; textbook Definition 1.2.1, p. 16)**.** The *degree* of a vertex v, written $d_G(v)$ or $d(v)$ in the textbook, is the number of edges incident to v, that is, the number of edges that contain v [9, Definition 1.2.1, p. 16]. This paper writes $\deg(n_i)$, to keep d for distance (Definition 2.48).

Remark 2.17. In a simple graph the degree equals the number of neighbours, since each neighbour is joined to n_i by exactly one edge. *[derived here]* The textbook places the definition of degree after the adjacency matrix, in Section 1.2, before the walks; this paper keeps that order.

Example 2.18. In B: $\deg(n_1) = 1$, $\deg(n_2) = 2$, $\deg(n_3) = 1$.

Non-example 2.19. $\deg(n_1)$ is not 2 in B, although two other houses can be reached from n_1. Only roads at n_1 count, and n_3 is reached by passing through n_2.

> **In plain words.** Adding up one row of the table counts the roads at that house. Adding up the whole table counts every road twice, once from each end.

Proposition 2.20 (Row sums and the total)**.** *For the adjacency matrix A of a simple undirected graph: (a) the sum of row i, and of column i, equals* $\deg(n_i)$*; (b) the sum of all entries equals $2|E|$. Part (b) is the textbook's Theorem 1.2.1 of Euler, $\sum_v d_G(v) = 2|E|$ [9, Theorem 1.2.1, p. 16], read through (a). Part (a) is stated, for graphs that may have loops, by Paramadevan and Sotheeswaran [7]. The proof here is [derived here].*

Proof. (a) The sum of row i is the number of j with $A_{ij} = 1$, that is, the number of neighbours of n_i. By Remark 2.17 this is $\deg(n_i)$. Column i gives the same by symmetry (Proposition 2.13). (b) Add (a) over all i. Each edge $n_i n_j$ is counted once in $\deg(n_i)$ and once in $\deg(n_j)$, so twice in all. □

Example 2.21. For the board matrix the row sums are $1, 2, 1$, the degrees of Example 2.18, and the sum of all entries is $4 = 2|E|$ with $|E| = 2$.

Non-example 2.22. The matrix of Non-example 2.15 has row sums $1, 2, 2$, but it is not an adjacency matrix, so part (a) does not apply to it.

2.2 Walks, trails, paths and cycles

The textbook uses the word "path" for a walk with no repeated edge (Definition 1.2.12) and "simple path" for a walk with no repeated vertex (Definition 1.2.13) [9]. English texts and the course convention use "path" for the second [3]. This paper uses *trail* for the first and *path* for the second, as announced in Section 1.3. Table 7 in Section 5 shows the three conventions side by side.

> **In plain words.** A trip is a list of houses. Each house on the list shares a road with the one before it. You may pass a house twice. You may drive a road twice. The length of a trip is the number of roads driven. A road driven twice counts twice. Staying at home is a trip of length zero. Books call a trip like this a walk.

Definition 2.23 (Walk; textbook Definition 1.2.10, p. 21)**.** A *walk of length k from u_0 to u_k*, also called a *(u_0, u_k)-walk*, is a sequence $u_0, e_1, u_1, \ldots, u_{k-1}, e_k, u_k$ of vertices and edges with $e_i = u_{i-1}u_i$ for $1 \le i \le k$ [9, Definition 1.2.10, p. 21]. Its *length* is k, the number of edges, each edge counted as many times as it occurs [9, remark after Definition 1.2.10, p. 22].

Remark 2.24. Vertices and edges may repeat in a walk, and the length counts every use of an edge [9, p. 22]. The only thing forbidden is a jump: two consecutive vertices must be adjacent. In a simple graph the edge e_i is fixed by its two ends (Remark 2.2), so a walk can be written by its vertices alone, $u_0, u_1, \ldots, u_k$, and two walks are different exactly when these lists differ at some position. *[derived here]* The textbook does not say in words that $k = 0$ is allowed. This paper allows it, because the textbook's own property $d(u, u) = 0$ (Definition 2.48) needs a length-0 trip. *[interpretation, qualified]*

Example 2.25. In B the list $1, 2, 1, 2, 3$ is a walk from n_1 to n_3 of length 4: its steps 12, 21, 12, 23 are all edges. Vertices 1 and 2 occur twice, and the edge 12 is used twice. The textbook's example is $v_1, v_3, v_2, v_4, v_3, v_2, v_4, v_3, v_5$ in G_{tb}, a walk of length 8 [9, p. 22].

Non-example 2.26. In B the list $1, 3$ is not a walk, because 13 is not an edge of B (the matrix has $A_{13} = 0$). It would be a walk of the drawing T. The textbook's non-example is $v_3, v_1, v_4, v_2, v_3, v_5$, which is not a walk of G_{tb} [9, p. 22].

> **In plain words.** A trip is closed when it ends at the house where it began. That is the only new rule. The trip may still pass houses and roads many times.

Definition 2.27 (Closed walk; textbook Definition 1.2.11, p. 22)**.** A (u_0, u_k)-walk is *closed* if $u_0 = u_k$ [9, Definition 1.2.11, p. 22].

Remark 2.28. A closed walk allows everything a walk allows; it forbids only a different start and end. A walk of length 0 is closed. In a simple graph a closed walk cannot have length 1 (that would be a loop), and one of length 2 must be u, w, u and so uses one edge twice. *[derived here]*

Example 2.29. In B the walks $2, 1, 2$ (length 2) and $2, 3, 2, 1, 2$ (length 4) are closed.

Non-example 2.30. In B the walk $1, 2, 3$ is not closed: it starts at 1 and ends at 3.

> **In plain words.** Now add one rule. Never drive the same road twice. You may still pass a house twice, if you return by another road. English books call such a trip a trail. Our textbook calls it a path. This paper says trail, so that the word path can keep one meaning.

Definition 2.31 (Trail; textbook Definition 1.2.12, p. 22, where it is called a "path")**.** A (u_0, u_k)-walk is a *trail* if its edges $u_0u_1, \ldots, u_{k-1}u_k$ are pairwise distinct. The textbook calls this a *path* [9, Definition 1.2.12, p. 22]; English usage calls it a trail [3].

Remark 2.32. A trail allows repeated vertices and forbids repeated edges. A vertex can recur in a trail only if the trip left it by one road and came back by another, so the trail contains a closed trail, a detour that comes back to the same house. In B every trail is also a path in the sense of Definition 2.35 (checked by script for lengths 0 to 4), so B cannot show the difference; the textbook's graph can. *[derived here]*

Example 2.33. In G_{tb} the list $v_1, v_3, v_2, v_4, v_3, v_5$ is a trail: its edges v_1v_3, v_3v_2, v_2v_4, v_4v_3, v_3v_5 are distinct, although v_3 occurs twice [9, p. 22].

Non-example 2.34. The walk $v_1, v_3, v_2, v_4, v_3, v_2, v_4, v_3, v_5$ of length 8 is not a trail: the edge v_3v_2 occurs twice [9, p. 22]. In B the walk $1, 2, 1, 2, 3$ is not a trail, since the edge 12 occurs twice.

> **In plain words.** Now add a stronger rule. Never pass the same house twice. Then you never drive a road twice either. A trip like this is a route. It goes straight from start to end. It has no detour in it. Our course calls it a path. The textbook calls it a simple path.

Definition 2.35 (Path; textbook Definition 1.2.13, p. 22, where it is called a "simple path")**.** A (u_0, u_k)-walk is a *path* if its vertices $u_0, \ldots, u_k$ are pairwise distinct. The textbook calls this a *simple path* and writes P_n for the simple path on n vertices [9, Definition 1.2.13, p. 22]. Its length is k.

Remark 2.36. A path forbids repeated vertices, hence also repeated edges (Proposition 2.39). It allows length 0 (a single vertex is a path from that vertex to itself) and length 1 (one edge). It does not allow a closed path of positive length, because the first and last vertex would be equal. The textbook defines a simple path as a trail with distinct vertices; by Proposition 2.39 this is the same as a walk with distinct vertices, so the two ways of saying it agree. That the *course* uses "path" in this sense is *[interpretation, qualified]*: it comes from the assignment text, and the board does not say.

Example 2.37. In B the list $1, 2, 3$ is a path from n_1 to n_3 of length 2. The textbook's example is v_1, v_3, v_4, v_5 in G_{tb}, a simple path [9, p. 22].

Non-example 2.38. In B the list $1, 2, 3, 2$ is a walk but not a path: vertex 2 occurs twice. In G_{tb} the list $v_1, v_3, v_2, v_4, v_3, v_5$ is a trail but not a simple path, since v_3 occurs twice [9, p. 22]. (In the printed text this sentence stands after Definition 1.2.14, not directly after 1.2.13; its content is about 1.2.13.)

> **In plain words.** A trip that never passes a house twice never drives a road twice. Here is why. Take two roads on the trip, an earlier one and a later one. The later road ends at a house that the earlier road does not touch. So the two roads are different.

Proposition 2.39 (A path is a trail)**.** *In a simple graph, a walk with pairwise distinct vertices has pairwise distinct edges. So every path is a trail. [derived here]*

Proof. Let $u_0, \ldots, u_k$ have pairwise distinct vertices, and take two steps with indices $a < b$, that is, the edges $u_{a-1}u_a$ and $u_{b-1}u_b$. The vertex u_b has an index larger than the indices $a-1$ and a, so it differs from both u_{a-1} and u_a. But u_b is an end of the second edge and is not an end of the first. So the two edges are different. (Grinberg leaves this statement as an exercise [3].) □

> **In plain words.** A cycle is a closed trip that never drives a road twice. It may pass a house twice on the way. An English book calls it a closed trail.

Definition 2.40 (Cycle; textbook Definition 1.2.14, p. 22)**.** A trail that is a closed walk is called a *closed path* or *cycle* [9, Definition 1.2.14, p. 22]. In the words of this paper, a *cycle* is a closed trail: a (u_0, u_k)-walk with $u_0 = u_k$ and pairwise distinct edges. Vertices may repeat.

Remark 2.41. The definition allows repeated vertices (a figure eight is a cycle) and forbids repeated edges and open walks. English usage differs: Grinberg's "cycle" needs distinct vertices and is the textbook's simple cycle [3]. The textbook is silent on whether a closed trail of length 0 is a cycle; this paper does not count it *[interpretation, qualified].*

Example 2.42. In G_{tb} the list $v_3, v_2, v_1, v_3, v_5, v_4, v_3$ is closed and has six distinct edges, so it is a cycle in this sense; it is not a simple cycle, since v_3 occurs three times [9, p. 22].

Non-example 2.43. In B the closed walk $1, 2, 1$ is not a cycle, because the edge 12 is used twice. In fact B has no cycle at all: a cycle is a closed trail, in a simple graph a closed trail of positive length needs at least three distinct edges (Remark 2.28), and B has only two edges. *[derived here]*

> **In plain words.** A simple cycle is stricter. It passes no house twice, except the start, which is also the end. It is a ring of houses, each joined by a road to the next.

Definition 2.44 (Simple cycle; textbook Definition 1.2.15, p. 22)**.** A cycle is *simple* if only its first and last vertices coincide. The simple cycle on n vertices is written C_n, with $n \geq 3$ [9, Definition 1.2.15, p. 22].

Remark 2.45. A simple cycle forbids any repeated vertex other than the closing one. The condition $n \geq 3$ is forced in a simple graph: a closed trail cannot have length 1 or 2 (Remark 2.28). *[derived here]*

Example 2.46. In G_{tb} the list v_1, v_3, v_4, v_2, v_1 is a simple cycle [9, p. 22]. In the board drawing T the list $1, 2, 3, 1$ is a simple cycle of length 3 (by script).

Non-example 2.47. The cycle $v_3, v_2, v_1, v_3, v_5, v_4, v_3$ of Example 2.42 is not simple, because v_3 occurs three times [9, p. 22].

> **In plain words.** The distance between two houses is the number of roads on the shortest trip between them. If no trip exists, the distance is called infinite. A house is at distance zero from itself. The distance there equals the distance back, because any trip can be driven backwards. Going by way of a third house is never shorter than the shortest trip.

Definition 2.48 (Distance; textbook Definition 1.2.16, pp. 22–23)**.** The *distance* $d(u, v) = d_G(u, v)$ between vertices u and v is the length of a shortest (u, v)-trail if at least one (u, v)-trail exists, and $+\infty$ otherwise [9, Definition 1.2.16, pp. 22–23]. (The textbook's word is "path", its Definition 1.2.12.)

Remark 2.49. The textbook lists three properties [9, p. 23]:

1. $d(u, v) \geq 0$, and $d(u, v) = 0$ if and only if $u = v$;
2. $d(u, v) = d(v, u)$;
3. $d(u, v) \leq d(u, w) + d(w, v)$.

So if any two vertices are joined by a trail, (V, d) is a metric space [9]; a *metric space* is just a set with a distance that satisfies these three properties, and the word is not used again. The definition allows the value $+\infty$, which is not an integer; this is how a missing trip shows up. It forbids negative values and any value that is not the length of an actual trip. The shortest trail, the shortest path and the shortest walk between two vertices have the same length, so the definition does not depend on whether "path" is read as trail or path (Appendix A.5). *[derived here]*

Example 2.50. In B: $d(1, 2) = 1$, $d(2, 3) = 1$, $d(1, 3) = 2$, and $d(i, i) = 0$ for each i (by script). The walk $1, 2, 1, 2, 3$ has length 4, but $d(1, 3)$ is 2, the shortest.

Non-example 2.51. The number 4 is not the distance from 1 to 3 in B, even though a walk of length 4 exists: distance is the minimum, not any length. In the graph D with vertices $1, 2, 3$ and the single edge 12 (Figure 3), $d(1, 3)$ is not a number at all; it is $+\infty$ (by script).

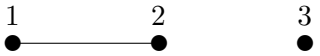


Figure 3: The graph D: three vertices and the single edge 12. Vertex 3 is isolated. It is the board graph B with the edge 23 removed.

2.3 From walks to paths

> **In plain words.** Suppose you can drive from one house to another. Then you can do so without passing any house twice. Look at your trip. If it passes some house twice, you went on a detour that came back to the same house. Cut the detour out. What is left is still a trip between the same two houses. It is shorter. Keep cutting detours until no house repeats. The trip that remains is never longer than the first one.

Lemma 2.52 (Every walk contains a path; textbook Lemma 1.2.1, p. 23, for $u \neq v$)**.** *Let u and v be vertices of a simple graph, equal or different. If there is a (u, v)-walk of length k, then there is a (u, v)-path of length at most k. All vertices and edges of this path are vertices and edges of the walk.*

The textbook states the lemma for $u \neq v$, as "from any (u, v)-walk one can extract a simple (u, v)-path", without a length bound, and proves it by induction on the length of the walk [9, Lemma 1.2.1, p. 23]. Grinberg's Corollary 2.9.7 covers $u = v$ and gives the length bound [3, Corollary 2.9.7, p. 39]. The case $u = v$ is needed here, because a closed walk such as $2, 1, 2$ in B has $u = v$; the path it contains is the single vertex, of length 0. The proof below is the paper's own *[derived here]*; it follows the textbook's method.

Proof. We prove by strong induction on k the statement $S(k)$: every (u, v)-walk of length k, for any vertices u and v, contains a (u, v)-path of length at most k.

Base case, $k = 0$. A walk of length 0 is a single vertex u, so $v = u$. A single vertex has no repeated vertex, so the walk is itself a path, of length $0 \leq 0$.

Induction hypothesis. Let $k \geq 1$ and assume $S(m)$ holds for every $m < k$.

Step. Let $u_0, u_1, \ldots, u_k$ be a (u, v)-walk, so $u_0 = u$ and $u_k = v$. Case 1: the vertices $u_0, \ldots, u_k$ are pairwise distinct. Then the walk is a path of length $k \leq k$. Case 2: some vertex occurs twice, so $u_i = u_j$ for some $i < j$. Form the list $W' = u_0, \ldots, u_i, u_{j+1}, \ldots, u_k$. If $j = k$, the part after u_i is empty and W' ends at u_i. We check that W' is a (u, v)-walk. (a) It starts at $u_0 = u$. (b) It ends at v: if $j < k$ its last vertex is $u_k = v$; if $j = k$ its last vertex is $u_i = u_j = u_k = v$. (c) Consecutive vertices are adjacent: before the cut and after the cut they are consecutive vertices of the old walk, and at the join the pair is $(u_i, u_{j+1}) = (u_j, u_{j+1})$, which is a step of the old walk. The length of W' is $i + (k - j)$, and since $i < j$ this is less than k. By the induction hypothesis W' contains a (u, v)-path of length at most that, hence at most k. Every vertex and edge of that path is one of W', and so one of the old walk. □

Why the proof is valid: each cut removes at least one step, since $j - i \geq 1$, so the length drops by at least 1 and the process stops. A walk that cannot be cut has no repeated vertex, which is the definition of a path.

Remark 2.53. The textbook's walk of length 8 in G_{tb} is $1, 3, 2, 4, 3, 2, 4, 3, 5$, written by the indices of its vertices, with positions counted from 0. Positions 1 and 4 both hold vertex 3 (so $u_1 = u_4 = 3$). Remove the part after position 1 up to and including position 4, that is, the entries $2, 4, 3$. What is left is $1, 3, 2, 4, 3, 5$, of length 5. In it, positions 1 and 4 again both hold vertex 3; removing the entries $2, 4, 3$ again leaves $1, 3, 5$, a path of length $2 \leq 8$. In B the walk $1, 2, 1, 2, 3$ has $u_0 = u_2 = 1$ and cuts to the path $1, 2, 3$; the closed walk $2, 1, 2$ cuts to the single vertex 2. (Checked by script, Appendix A.7.)

> **In plain words.** A trip that never repeats a house can pass each house at most once. So when the number of houses is fixed, such a trip is short. It uses at most one road fewer than the number of houses.

Corollary 2.54 (Length bound)**.** *In a graph with N vertices every path has at most $N - 1$ edges. Hence, if there is a walk from u to v, there is a walk from u to v of length at most $N - 1$. [derived here]*

Proof. A path of length k lists $k + 1$ pairwise distinct vertices, and there are only N vertices, so $k + 1 \leq N$, that is $k \leq N - 1$. For the second part, Lemma 2.52 gives a path inside the walk, of length at most that of the walk and at most $N - 1$; a path is itself a walk. Grinberg gives the same argument [3, Section 2.9.4, p. 40]. □

For B, $N = 3$, so a path has at most 2 edges; the longest path in B has 2 edges (by script).

2.4 Connected graphs and components

> **In plain words.** Sometimes you can reach every house from every other house. Then the village is called connected. If one house is cut off, the village is not connected.

Definition 2.55 (Connected; textbook Definition 2.1.1, p. 48; board, last line)**.** A graph G is *connected* if for any two vertices u and v of G there is a (u, v)-path in G [9, Definition 2.1.1, p. 48]. The board says the same: "Graph is called connected if for any n_i, n_j there exists a path between nodes" [4, 5].

Remark 2.56. In the textbook the word "path" here is Definition 1.2.12, a trail; the board does not say which meaning it uses. The next proposition shows that the choice does not change

which graphs are connected. "Any two vertices" includes $u = v$; for $u = v$ the one-vertex walk is a (u, u)-path, so these pairs are always satisfied, and reading the definition as "any two different vertices" changes nothing. *[interpretation; the textbook does not discuss $u = v$]* The definition forbids exactly one thing: a pair of vertices with no trip between them.

Example 2.57. The board graph B is connected: the pair $1, 3$ is joined by the path $1, 2, 3$, and the pairs $1, 2$ and $2, 3$ by one edge each. The textbook's examples of connected graphs are the complete graphs, the complete bipartite graphs, and the three graphs of its Fig. 2.1.1 [9, p. 48].

Non-example 2.58. The graph D (Figure 3) is not connected: no walk starts at vertex 3 and ends elsewhere, so the pair $1, 3$ has no path. The textbook's non-example is the graph of its Fig. 2.1.2, which has four components [9, p. 49].

> **In plain words.** It does not matter which kind of trip we ask for. If any trip links two houses, a route links them. Every route is a trail. Every trail is a trip. So all three questions have the same answer.

Proposition 2.59 (Three readings of "connected" agree)**.** *For vertices u, v of a simple graph the following are equivalent: (a) there is a (u, v)-walk; (b) there is a (u, v)-trail; (c) there is a (u, v)-path. Hence G is connected in the textbook's sense (trails) if and only if every two vertices are joined by a path, if and only if every two vertices are joined by a walk. [derived here from Lemma 1.2.1 and Corollary 2.9.7]*

Proof. (c)⇒(b): every path is a trail (Proposition 2.39). (b)⇒(a): every trail is a walk, by definition. (a)⇒(c): Lemma 2.52. Grinberg states the equivalence of (a) and (c) as his Proposition 2.9.10 [3, p. 42]. □

> **In plain words.** A village that is not connected falls into pieces. Inside one piece every house can reach every other house. No road joins two different pieces. Each piece is called a component. A connected village has exactly one piece.

Definition 2.60 (Connected component; textbook, p. 49, paragraph before Remark 2.1.1)**.** On the vertex set V write $u \sim v$ if there is a (u, v)-path in G. This relation is *reflexive* (every vertex is related to itself), *symmetric* (if $u \sim v$ then $v \sim u$) and *transitive* (if $u \sim v$ and $v \sim w$ then $u \sim w$) [9, p. 49]. Here is why, reading "path" as walk, which is allowed by Proposition 2.59: the one-vertex walk gives reflexivity, reversing a walk gives symmetry, and joining a (u, v)-walk to a (v, w)-walk gives a (u, w)-walk, which gives transitivity *[derived here]*. Hence V splits into *classes* $V_1, \ldots, V_p$: groups of vertices such that each vertex lies in exactly one group, and two vertices lie in the same group exactly when they are related. The subgraphs $G_j = G[V_j]$ (the vertices of V_j with every edge of G that has both ends in V_j) are the *connected components* of G [9, p. 49]. The textbook gives this in prose, without a definition number.

Remark 2.61. A component is a whole class: a vertex reachable from a vertex of V_j lies in V_j. The textbook's Remark 2.1.1 states that G is connected if and only if it has exactly one connectivity component [9, Remark 2.1.1, p. 49]. By Proposition 2.59 the relation $\sim$ is the same whether "path" means trail, path or walk.

Example 2.62. The graph D has the two components $G[\{1, 2\}]$ and $G[\{3\}]$; the board graph B has the single component $G[\{1, 2, 3\}]$ (checked by breadth-first search in the script, Appendix A.7).

Non-example 2.63. $G[\{1\}]$ is not a component of D, although it is a connected subgraph: vertex 2 can be reached from vertex 1, so 2 lies in the same class as 1. A component is a whole class, not any connected piece. *[derived here]*

3 Answer 1: what the entry of a power of A counts

Question 1 asks what $(A^k)_{ij}$ represents. The answer is a count of walks (Definition 2.23), and the proof needs only the definition of the matrix product, so that product comes first.

3.1 Matrix products and powers

> **In plain words.** To multiply two tables, take a row from the first and a column from the second. Multiply the numbers in pairs. Then add the results. Do this for every row and every column to build the new table. Multiply the road table by itself again and again, and you get its powers.

Definition 3.1 (Matrix product and power)**.** Let X and Y be $N \times N$ matrices. Their *product* XY is the $N \times N$ matrix with

$$(XY)_{ij} = \sum_{r=1}^{N} X_{ir}\, Y_{rj}. \tag{1}$$

The *powers* of a square matrix A are $A^1 = A$ and $A^k = A^{k-1}A$ for $k \geq 2$.

Remark 3.2. Formula (1) multiplies row i of X with column j of Y entry by entry and adds. A power is the repeated product of the whole matrix, not a power of each entry. This paper uses exponents $k \geq 1$ only.

Example 3.3. For the board matrix, $(A^2)_{22} = A_{21}A_{12} + A_{22}A_{22} + A_{23}A_{32} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2$ (by script, Appendix A.7).

Non-example 3.4. A^2 is not the matrix of squared entries. Squaring each entry of the board matrix gives back A, whose entry $(1, 2)$ is 1, while $(A^2)_{12} = 0$.

3.2 The walk-counting theorem

> **In plain words.** Take the table of roads and multiply it by itself. Each number in the new table counts trips of two roads. The trips run from the row house to the column house. Multiply again and you count trips of three roads. The count includes trips that turn back. So the table counts trips, not routes.

Theorem 3.5 (Walk counting)**.** *Let A be the adjacency matrix of a simple undirected graph G with vertices $n_1, \ldots, n_N$, and let $k \geq 1$ be an integer. Then $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j, for all i, j.*

The statement is Theorem 1.1.3 of the University of Galway notes, for exactly this setting [2, Theorem 1.1.3]. Grinberg, Lehman, Leighton and Meyer, and Bai state it too, in more general settings [3, Theorem 4.5.10, pp. 120–121][6, Corollary 9.3.3][1]. The textbook does not state the theorem in the pages checked *[qualified]*, so the proof below is written out in full. It follows the standard route: induction on k, splitting the walks by their second-last vertex [2, 3]. The write-up is *[derived here]*. One journal paper states a different version, with "paths" in place of walks; it is false, and Section 3.4 shows why.

Remark 3.6. The theorem allows $i = j$, which counts closed walks, and it counts every walk once, with repeated vertices and edges allowed. It forbids reading the number as a count of paths (Proposition 3.10). It says nothing about walks of other lengths: $(A^2)_{12} = 0$ on the board graph does not say that n_1 and n_2 are unconnected.

Example 3.7. On the board graph B, $(A^2)_{13} = 1$: the only walk of length 2 from n_1 to n_3 is $1, 2, 3$. And $(A^3)_{12} = 2$: the walks of length 3 from n_1 to n_2 are $1, 2, 1, 2$ and $1, 2, 3, 2$. All entries are written out in Section 6.

Non-example 3.8. $(A^2)_{12} = 0$ on B does not mean that no walk joins n_1 and n_2. It means only that no walk of length *exactly* 2 does; walks of length 1 and of length 3 exist.

Proof. We use induction on k. Write $W_k(i,j)$ for the number of walks of length k from n_i to n_j. We prove $W_k(i,j) = (A^k)_{ij}$ for all i, j, for each $k \geq 1$. In a simple graph a walk is determined by its list of vertices (Remark 2.24). So $W_k(i,j)$ is the number of lists $(w_0, \ldots, w_k)$ of vertices with $w_0 = n_i$, $w_k = n_j$, and w_{t-1} adjacent to w_t for each t from 1 to k.

Base case, $k = 1$. A walk of length 1 from n_i to n_j is a list (n_i, n_j) with n_i and n_j adjacent. There is exactly one such list if they are adjacent and none otherwise. So $W_1(i,j) = A_{ij} = (A^1)_{ij}$.

Induction hypothesis. Let $k \geq 2$ and suppose $W_{k-1}(i,j) = (A^{k-1})_{ij}$ for all i, j.

Induction step. Fix i and j. Every walk $(w_0, \ldots, w_k)$ of length k from n_i to n_j has a second-last vertex w_{k-1}. Because $k \geq 2$, deleting the last vertex leaves $w_0, \ldots, w_{k-1}$, which still has length $k - 1 \geq 1$, so the hypothesis applies to it. (For $k = 1$ nothing would remain to count; this is why $k = 1$ is the separate base case.) Let $w_{k-1} = n_r$, where r is one of $1, \ldots, N$. Sort the walks into N classes by the value of r. The classes are disjoint, since each walk has exactly one second-last vertex, and together they contain every walk. So

$$W_k(i,j) = \sum_{r=1}^{N} \big(\text{number of walks of length } k \text{ from } n_i \text{ to } n_j \text{ with second-last vertex } n_r\big).$$

We claim that the number in class r is $W_{k-1}(i,r)\, A_{rj}$.

- If n_r and n_j are not adjacent, then $A_{rj} = 0$. No walk can have n_r followed by n_j, so class r is empty, and its count 0 equals $W_{k-1}(i,r) \cdot 0$.
- If n_r and n_j are adjacent, then $A_{rj} = 1$. Delete the last vertex of a walk in class r. What remains is a list $w_0, \ldots, w_{k-1}$ with $w_0 = n_i$, $w_{k-1} = n_r$ and consecutive vertices adjacent, so it is a walk of length $k-1$ from n_i to n_r. Conversely, take any walk of length $k-1$ from n_i to n_r and append n_j. The new pair n_r, n_j is adjacent, so the result is a walk of length k from n_i to n_j with second-last vertex n_r. Deleting and appending undo each other, so they match the walks in class r one to one with the walks of length $k-1$ from n_i to n_r. The count is $W_{k-1}(i,r) = W_{k-1}(i,r) \cdot 1$.

Therefore $W_k(i,j) = \sum_r W_{k-1}(i,r)\, A_{rj}$. By the induction hypothesis $W_{k-1}(i,r) = (A^{k-1})_{ir}$, so

$$W_k(i,j) = \sum_{r=1}^{N} (A^{k-1})_{ir}\, A_{rj} = (A^{k-1}A)_{ij} = (A^k)_{ij},$$

by formula (1) and the definition of powers. This proves the statement for k, and so, by induction, for every $k \geq 1$. □

Why the argument is valid. Two facts carry it: the sum over the second-last vertex covers each walk exactly once, and the matching in the case $A_{rj} = 1$ is one to one. Simplicity is used twice: to identify a walk with its vertex list, and so that A_{rj}, which is the number of edges between n_r and n_j, is 0 or 1. Undirectedness is not used inside the induction; it only makes A symmetric. *[interpretation]* The proof is checked, not replaced, by a finite test. A *case* is one choice of a graph, an exponent k and a pair i, j; in each case a script counted the walks by brute force and compared the count with the entry $(A^k)_{ij}$. For every labelled simple graph on $2, 3, 4, 5$ vertices (2, 8, 64, 1024 graphs), with k up to 5 for $N \leq 4$ and up to 3 for $N = 5$, there are $2 \cdot 5 \cdot 4 = 40$, $8 \cdot 5 \cdot 9 = 360$, $64 \cdot 5 \cdot 16 = 5120$ and $1024 \cdot 3 \cdot 25 = 76800$ cases, and all are equal (Appendix A.6). That test is a check, not a proof.

3.3 Consequences of the theorem

In plain words. A count is never negative and always a whole number. The table of trips looks the same when flipped over its diagonal. A positive number means that at least one such trip exists. For trips of two roads, the diagonal counts the roads at a house.

Corollary 3.9. *For a simple undirected graph and $k \geq 1$:*

(a) every entry of A^k is a non-negative integer, so a sum of such powers has a positive (i, j) entry if and only if at least one summand does;

(b) $(A^k)_{ij} > 0$ if and only if there is a walk of length exactly k from n_i to n_j;

(c) A^k is symmetric;

(d) $(A^2)_{ii} = \deg(n_i)$, so the sum of the diagonal entries of A^2 equals $2|E|$.

[derived here]

Proof. Parts (a) and (b) follow because by Theorem 3.5 each entry is a count. For (c), $(A^k)^T = (A^T)^k = A^k$, where the *transpose* X^T of a matrix X is the matrix with its rows and columns exchanged, and $A^T = A$ because A is symmetric (Proposition 2.13); alternatively, reversing a walk matches the walks from n_i to n_j with the walks from n_j to n_i. For (d), the walks of length 2 from n_i to n_i are the lists n_i, n_r, n_i with n_r a neighbour of n_i, one for each neighbour, and the number of neighbours is $\deg(n_i)$ (Remark 2.17). Summing over i and using part (b) of Proposition 2.20 gives $2|E|$. The University of Galway notes state (d) in a remark after their theorem [2]. □

On the board graph the diagonal of A^2 is $(1, 2, 1)$, the degrees, and its sum is $4 = 2|E|$ (script).

3.4 The warning: walks are not paths

In plain words. A trip may turn back and pass the same house twice. A route never passes a house twice. The table counts trips, so it also counts the trips that turn back. A house with at least one neighbour always has a two-road trip that goes out and comes home. That trip is no route.

Proposition 3.10 (The entry is not a count of paths)**.** *Let G be a simple undirected graph.*

(a) If n_i has at least one neighbour, then $(A^2)_{ii} = \deg(n_i) > 0$, yet there is no path, and no trail, of length 2 from n_i to n_i.

(b) On the board graph B, $(A^3)_{12} = 2$, although the only path from n_1 to n_2 has length 1. More generally, on B (and not for every graph) $A^{k+2} = 2A^k$ for $k \geq 1$, so the entries grow with k while the number of paths does not.

[derived here]

Proof. (a) By Corollary 3.9(d), $(A^2)_{ii} = \deg(n_i)$, which is positive. The only walks of length 2 from n_i to n_i are n_i, n_r, n_i. They repeat the vertex n_i, so they are not paths (Definition 2.35); a path from a vertex to itself has pairwise distinct vertices, so its length is 0. They repeat the edge $n_i n_r$, so they are not trails either (Definition 2.31). The textbook's own word "path" (Definition 1.2.12) means a trail, so the entry is not a count of the textbook's paths either. (b) On B the walks of length 3 from n_1 to n_2 are $1, 2, 1, 2$ and $1, 2, 3, 2$. Each repeats a vertex and an edge, so neither is a path or a trail. The only path from n_1 to n_2 is the single edge $1, 2$. For the growth, $A^3 = 2A$ (script) and multiplying by A^{k-1} gives $A^{k+2} = 2A^k$. □

The sources say the same in words: Bai remarks that the entries count walks and not paths [1], and Grinberg's Example 2.9.2 shows a walk that is not a path [3, p. 37] . Table 5 gives the counts from the script for $k = 1, \ldots, 5$: the walk counts $1, 0, 2, 0, 4$ against the path counts $1, 0, 0, 0, 0$.

Table 5: Walks of length k from n_1 to n_2 on the board graph B, and how many of them are paths and trails (by script). The walk counts equal the entries $(A^k)_{12}$ of Section 6.

k	walks	of which paths	of which trails
1	1	1	1
2	0	0	0
3	2	0	0
4	0	0	0
5	4	0	0

Paramadevan and Sotheeswaran's Theorem 1 reads: "the number of different paths of length l between the i-th and j-th vertices of G is equal to the (i, j)-th entry of A_G^l" [7]. As written this is false for any definition of path that forbids repeated vertices or repeated edges, by Proposition 3.10(a); their own property 7 says walks [7]. So the sources disagree, and the paper follows the walk version, which four sources state and which was proved above.

What a positive entry *does* guarantee is a walk of length k, and then, by Lemma 2.52, a path of length at most k. This is why the warning does not harm the connectivity test of Section 4; but the value of the entry is a count of walks, not of paths. *[derived here]*

4 Answer 2: testing connectedness from A alone

Question 2 asks how to know from A whether the graph is connected (Definition 2.55). The idea is to turn the question about trips into a question about counts. By Theorem 3.5, the entries of the powers of A count walks. By Lemma 2.52, a walk between two vertices contains a path between them, and by Corollary 2.54 that path is short. So it is enough to look at walks of length at most $N - 1$.

4.1 The running total S

> **In plain words.** We need a name for the running total. It adds the counts of trips with one road, with two roads, and so on. We stop one step short of the number of houses. The whole check uses this one total.

Definition 4.1 (The matrix S; new notation, not from the textbook)**.** For a graph with $N \geq 2$ vertices, let

$$S = A + A^2 + \cdots + A^{N-1}, \tag{2}$$

the sum of the first $N - 1$ powers of A, added entry by entry.

Remark 4.2. By Theorem 3.5, S_{ij} is the number of walks from n_i to n_j whose length is between 1 and $N - 1$. Hence every entry of S is a non-negative integer. The diagonal of S counts closed walks and plays no part in the test (Remark 4.11). *[derived here]*

Example 4.3. For the board graph B, $N = 3$, so $S = A + A^2$:

$$S = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Non-example 4.4. For $N = 3$, neither A alone nor A^2 alone is S. The matrix A^2 of B has the zero entry $(1, 2)$ although n_1 and n_2 are joined by an edge. The matrix A has the zero entry $(1, 3)$ although n_1 and n_3 are connected through n_2. Either one alone would wrongly report "not connected". *[derived here]*

4.2 The test and its proof

> **In plain words.** We want a quick check for a village. We only have the table of shared roads. From the table we can count trips. Count the trips with one road, with two roads, and so on. Add the counts for each pair of houses. If no pair of different houses has a total of zero, the village is connected. We never need trips longer than one road fewer than the number of houses. A trip that wastes time can always be shortened. So longer trips tell us nothing new.

Proposition 4.5 (One pair at a time)**.** *Let G have $N \geq 2$ vertices and let $i \neq j$. The following are equivalent.*

(i) There is a path from n_i to n_j (distinct vertices; by Proposition 2.59 this is the same as asking for a trail or a walk).

(ii) There is an integer k with $1 \leq k \leq N - 1$ and $(A^k)_{ij} > 0$.

(iii) $S_{ij} > 0$.

Proof. (i)⇒(ii). Let P be a path from n_i to n_j, of length k. By Corollary 2.54, $k \leq N - 1$. Because $n_i \neq n_j$, a walk of length 0 cannot join them, so $k \geq 1$. P is a walk of length k from n_i to n_j, so by Theorem 3.5 the number $(A^k)_{ij}$ is at least 1.

(ii)⇒(iii). $S_{ij} = A_{ij} + (A^2)_{ij} + \cdots + (A^{N-1})_{ij}$. Each term is a count, hence a non-negative integer (Corollary 3.9(a)). A sum of non-negative numbers is at least as large as any one of its terms. The term with the index k of (ii) is positive, so $S_{ij} > 0$.

(iii)⇒(ii). If $S_{ij} > 0$, a sum of non-negative terms is positive, so at least one term $(A^k)_{ij}$ with $1 \leq k \leq N - 1$ is positive.

(ii)⇒(i). If $(A^k)_{ij} > 0$ with $k \geq 1$, then by Theorem 3.5 there is at least one walk of length k from n_i to n_j. Since $n_i \neq n_j$, Lemma 2.52 gives a path from n_i to n_j. □

> **In plain words.** Here is the answer in short. Pick any two different houses. Look for a trip between them. It never needs more roads than the number of houses, less one. The village is connected when every such pair has one. The table of counts lets us read this off.

Theorem 4.6 (The connectivity test)**.** *Let G be a graph with $N \geq 2$ vertices and adjacency matrix A. Then G is connected if and only if every entry off the diagonal of $S = A + A^2 + \cdots + A^{N-1}$ is positive.*

The statement appears in Paramadevan and Sotheeswaran's Theorem 2 [7] and in Weisstein's entry on connected graphs [11]. The proof in the first is incomplete (Section 1.6 quotes it), so the paper proves the test itself and cites these sources for the statement only. The textbook does not state it in the pages checked *[qualified]*. The proof is *[derived here]*.

Remark 4.7. The test allows any graph with $N \geq 2$ vertices, connected or not. It looks only at entries with $i \neq j$. It forbids requiring positive diagonal entries (Remark 4.11) and forbids stopping at A^{N-2} (Proposition 4.17).

Example 4.8. The board graph B has S as in Example 4.3: no zero entry, so B is connected, which agrees with Example 2.57.

Non-example 4.9. The graph D (one edge 12) has the matrix S written out in Section 6, with zeros at $(1,3)$, $(2,3)$, $(3,1)$, $(3,2)$. So D is not connected, which agrees with Non-example 2.58.

Proof. By Definition 2.55 and Proposition 2.59, G is connected if and only if every two vertices are joined by a path. For a pair with $u = v$ this is always true, by the one-vertex path (Remark 2.56). For a pair n_i, n_j with $i \neq j$ it is statement (i) of Proposition 4.5. By Proposition 4.5 that is equivalent to $S_{ij} > 0$. So G is connected if and only if $S_{ij} > 0$ for every $i \neq j$. □

Why the argument is valid. It uses three results and nothing else: the walk count (Theorem 3.5), the walk-to-path lemma (Lemma 2.52) with its length bound (Corollary 2.54), and the fact that a count is a non-negative integer, so a sum is positive exactly when a term is. The last step would fail with subtraction or arithmetic modulo a number, which can cancel positive terms. The exponent $N - 1$ is the largest one needed because a path has at most $N - 1$ edges; adding higher powers cannot change which off-diagonal entries are positive.

> **In plain words.** The same counts also tell us the pieces of a village. Two different houses are in the same piece exactly when some trip joins them. To find the distance, count the trips of one road, then of two roads, and so on. The first number of roads that has a trip is the distance.

Corollary 4.10 (Components and distances from the powers)**.** *For any graph with $N \geq 2$ vertices, connected or not, and $i \neq j$: (a) $S_{ij} > 0$ if and only if n_i and n_j lie in the same component. (b) If they do, $d(n_i, n_j)$ is the smallest $k \geq 1$ with $(A^k)_{ij} > 0$; if they do not, no power has a positive (i, j) entry and $d(n_i, n_j) = +\infty$. [derived here]*

Proof. (a) Proposition 4.5 never uses that G is connected. Its statement (i) says $n_i \sim n_j$, which is the definition of lying in one class, that is in one component (Definition 2.60). (b) Let k_0 be the smallest $k \geq 1$ with $(A^k)_{ij} > 0$. By Theorem 3.5 a walk of length k_0 exists and no shorter walk does. By Lemma 2.52 a path of length at most k_0 exists; being a walk, it has length at least k_0. So the shortest walk has length k_0, and the distance equals the length of the shortest walk (Appendix A.5). If no power has a positive entry, there is no walk and the distance is $+\infty$. □

The distance statement is also made by Lehman, Leighton and Meyer, and by Paramadevan and Sotheeswaran in their Theorem 3 [6, 7]. The entry S_{ij} tells whether n_i and n_j are connected, not which path joins them.

Remark 4.11 (The diagonal, and the cases $N = 1$ and $N = 2$). For $N = 2$ and one edge, $S = A$ has zeros on the diagonal although the graph is connected (by script: $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$). So the diagonal of S cannot be required to be positive. For $N \geq 3$ and G connected the diagonal of S is positive anyway, because every vertex has a neighbour (an isolated vertex would have no path to the other $N-1$ vertices) and $(A^2)_{ii} = \deg(n_i)$ (Corollary 3.9(d)); the test does not need this. For $N = 1$ there are no pairs of different vertices and the graph is connected; the sum S is empty, so the test is stated for $N \geq 2$. *[derived here]*

4.3 The test as a procedure

> **In plain words.** Here is the check, step by step. First write the table of the village. Then multiply the table by itself again and again. After each step, add the new counts to a running total. Stop after one step fewer than the number of houses. Look at every pair of different houses. If any total is zero, those two houses cannot reach each other. If no total is zero, everyone can reach everyone.

Test 4.12 (Matrix test for connectedness)**.** *Input:* the adjacency matrix A of a graph with $N \geq 2$ vertices.

1. Set $P = A$ and $S = A$.
2. For $k = 2, 3, \ldots, N-1$: replace P by PA (now $P = A^k$) and add P to S.
3. Look at the entries S_{ij} with $i \neq j$. If one of them is 0, report "not connected". If none is 0, report "connected".

Remark 4.13. Correctness is Theorem 4.6. The loop makes $N-2$ matrix products, the last one producing A^{N-1}. A zero at position (i, j) means by Corollary 4.10 that n_i and n_j are in different components, so the procedure also lists the pieces. The test forbids stopping early: it needs every power up to $N-1$ (Proposition 4.17).

Example 4.14. On B ($N = 3$) the loop runs once ($k = 2$): $P = A^2$, $S = A + A^2$, no off-diagonal zero, so "connected". The matrices are in Section 6.

Non-example 4.15. Reporting after $k = 1$ only (the table A itself) is not the test: on B it would report "not connected" because $A_{13} = 0$.

4.4 An equivalent form with one table

> **In plain words.** There is a second form of the check that uses one table only. Let a trip wait at a house for a step. Then count trips of one fixed length. Waiting lets short trips count too. The village is connected exactly when no pair of houses has a count of zero.

The table that makes a trip wait is the identity matrix I, which has ones on the diagonal and zeros elsewhere: "stay at the same house" is one allowed step, just as "drive one road" is the step given by A. When $(I + A)$ is multiplied by itself m times, each of the m factors offers a choice between waiting (I) and driving one road (A). A choice with k drives and $m - k$ waits gives the product A^k (because $IX = X$ for every matrix X), and there are $\binom{m}{k}$ such choices. So

$$(I + A)^m = \binom{m}{0} I + \binom{m}{1} A + \binom{m}{2} A^2 + \cdots + \binom{m}{m} A^m, \tag{3}$$

with $A^0 = I$. Every coefficient is a positive integer. Appendix A.4 gives the argument in detail.

> **In plain words.** So one table is enough. Take the table that allows waiting and raise it to a high enough power. A zero anywhere means that two houses cannot reach each other. Otherwise every house can reach every other house.

Proposition 4.16 (The form with $I + A$)**.** *Let G have $N \geq 2$ vertices, let I be the $N \times N$ identity matrix, and let $m \geq N - 1$ be an integer. Then (a) for $i \neq j$, $\big((I + A)^m\big)_{ij} > 0$ if and only if $S_{ij} > 0$; (b) every diagonal entry of $(I + A)^m$ is at least 1. Hence G is connected if and only if $(I + A)^{N-1}$ has no zero entry. [derived here; proof in Appendix A.4]*

Bai states the case $m = N - 1$ as his Theorem 2, with a proof sketch that counts "paths" where walks are meant [1]. That statement is therefore *[qualified]*; the proof in the appendix replaces the sketch. For the board graph and for D the matrices $(I + A)^2$ are in Section 6, and the zeros fall in the same places as the zeros of S.

4.5 Why $N-1$ and not less

In plain words. Take four houses in a row. To go from the first house to the last you must pass every house in between. That takes three roads. If we counted only trips of up to two roads, we would miss this pair. We would say the village is cut in two, and we would be wrong. The same idea works for any row of houses.

Proposition 4.17 ($N-1$ cannot be lowered to $N-2$)**.** *For every $N \geq 3$ there is a connected graph with N vertices for which the sum $A + \cdots + A^{N-2}$ has a zero entry off the diagonal. [derived here]*

Proof. Take the graph with edges 1–2, 2–3, …, $(N-1)$–N, a row of N houses; it is connected. Each step of a walk changes the vertex number by exactly 1. To go from vertex 1 to vertex N the number must increase by $N-1$, so at least $N-1$ steps are needed. A walk of length $k < N-1$ from 1 to N does not exist, so by Theorem 3.5 $(A^k)_{1N} = 0$ for $1 \leq k \leq N-2$. Hence the $(1, N)$ entry of $A + \cdots + A^{N-2}$ is 0, although the graph is connected. □

For $N = 3$ the graph is the board graph and the sum $A+\cdots+A^{N-2}$ is A itself, with $A_{13} = 0$. For $N = 4$ (Figure 4) every matrix is written out below; the sum of the first $N-2 = 2$ powers has zeros at $(1, 4)$ and $(4, 1)$, and the sum of the first $N-1 = 3$ powers has none.



Figure 4: Four houses in a row, the graph used in Example 4.18.

Example 4.18 (The bound $N-1$ is needed; four houses in a row)**.** With $N = 4$, the matrices from the script are

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad A^2 = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix},$$

$$A^3 = \begin{pmatrix} 0 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 0 & 2 \\ 1 & 0 & 2 & 0 \end{pmatrix}, \qquad A + A^2 = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix},$$

$$A + A^2 + A^3 = \begin{pmatrix} 1 & 3 & 1 & 1 \\ 3 & 2 & 4 & 1 \\ 1 & 4 & 2 & 3 \\ 1 & 1 & 3 & 1 \end{pmatrix}.$$

With $N-2 = 2$ powers the $(1, 4)$ entry is 0, so the test would wrongly say "not connected". With $N-1 = 3$ powers every entry is positive.

Using only the single power A^{N-1} is also wrong: on the board graph A^2 has the zero entry $(1, 2)$.

Remark 4.19 (Which graphs the $N-2$ version gets wrong). The test with only $N-2$ powers is wrong on exactly the labelled rows of N houses, that is, the path graphs with the vertices numbered in any order. There are $N!/2$ of them for $N \geq 2$ ($N!$ orders along the row, each row counted twice because it can be read from either end): $1, 3, 12, 60, 360$ for $N = 2, \ldots, 6$. These are the numbers in the last column of Table 10, which the script counted by brute force. *[derived here]* Reason: the $N-2$ version can only err by saying "not connected" for a connected graph, because a positive entry always comes from a real walk. By Corollary 4.10(b) it says so exactly

Table 6: The row of N houses, $N = 3, \ldots, 8$ (by script). The first power with a non-zero $(1, N)$ entry is always $N - 1$.

N	smallest k with $(A^k)_{1N} > 0$	$(1, N)$ entry of $A + \cdots + A^{N-2}$	$(1, N)$ entry of $A + \cdots + A^{N-1}$
3	2	0	1
4	3	0	1
5	4	0	1
6	5	0	1
7	6	0	1
8	7	0	1

when some pair of vertices has distance $N - 1$. A shortest path between two such vertices passes through all N vertices, and a shortest path has no edge between vertices that are not neighbours on it, since that edge would give a shorter trip. So the graph is that row of houses. Conversely, Proposition 4.17 shows that a row fails. For $N = 2$ the sum of $N - 2 = 0$ powers is empty, and an empty sum is the zero matrix, so the test says "not connected" on the one connected graph.

4.6 Breadth-first search, the alternative

> **In plain words.** There is another way, and it needs no table of counts. Stand at one house and mark it. Then mark every house next to it. Then mark every house next to those. Keep going until no new house appears. If every house is marked, the village is connected.

Breadth-first search (BFS) from a start vertex s is this procedure.

1. Mark s and put it on a waiting list, called a *queue*: vertices leave it in the order in which they entered.
2. Take the first vertex off the list. Mark every neighbour of it that is not yet marked (these vertices are now *discovered*) and put them at the end of the list.
3. Repeat step 2 until the list is empty.
4. Report "connected" if every vertex is marked, and "not connected" otherwise.

The correctness statement in the source is that for every vertex v there is a walk from s to v if and only if v is discovered when the search ends, so G is connected if and only if every vertex is discovered when BFS starts from any single vertex s [10]. When the graph is stored as adjacency lists (for each vertex, the list of its neighbours), the source gives a running time at most a fixed multiple of $N + |E|$ [10]. BFS and the matrix test agree on all labelled graphs with $2, 3, 4, 5, 6$ vertices (2, 8, 64, 1024, 32768 graphs; a check, not a proof, Appendix A.6), and BFS also lists the components directly. The two methods are compared for cost in Section 7.

5 Answer 3: the definition of a path between n_i and n_j

Question 3 asks for the definition of a path between two vertices n_i and n_j. The answer is Definition 2.35, restated for a pair of vertices in Definition 5.1 below: in the course, a path is a walk with no repeated vertex, which is the textbook's Definition 1.2.13 [9, Definition 1.2.13, p. 22]. This section adds only the form for a pair of vertices and the comparison of the three naming conventions. The board uses the word in its definition of "connected" and does not say which meaning it has [4, 5]; the assignment text says that in the course a path has no repeated vertices. That the course uses this meaning is *[interpretation, qualified]*, because it comes from the assignment text and not from the board.

In plain words. A path between two houses is a route from one to the other. It never passes a house twice, so it has no detour. If the two houses are the same, the route has no roads at all. It is just the one house.

Definition 5.1 (Path between n_i and n_j, course convention)**.** Let n_i and n_j be vertices of G. A *path between n_i and n_j* is a path in the sense of Definition 2.35 whose first vertex is n_i and whose last vertex is n_j: a walk $w_0, w_1, \ldots, w_k$ with $w_0 = n_i$, $w_k = n_j$ and pairwise distinct vertices. Its length is k. *[interpretation, qualified]*

This is a simple path in the textbook's sense and a path in English usage [9, 3]; Grinberg phrases the pair form as "a path from p to q is a path that starts at p and ends at q" [3, Definition 2.9.1(d), p. 36].

Remark 5.2. The definition forbids repeated vertices, and so repeated edges (Proposition 2.39). For $i = j$ the only such path is the single vertex n_i, of length 0 (this uses the reading of length-0 walks in Section 1.3, item 7). For $i \neq j$ the length is at least 1 and, by Corollary 2.54, at most $N-1$. In the board's sentence "there exists a path between nodes", the words mean that at least one such object exists. By Proposition 2.59 that statement has the same truth value whether "path" means walk, trail or path in the sense above. What changes with the convention is how many such objects there are (Remark 5.5).

Example 5.3. Example 2.37 gives the path $1, 2, 3$ between n_1 and n_3 in B. It is the only one: every path has at most $N - 1 = 2$ edges (Corollary 2.54), and the enumeration of walks up to length 4 finds no other (by script). Between n_1 and n_1 the only path is the single vertex 1.

Non-example 5.4. The walk $1, 2, 1, 2, 3$ of Example 2.25 joins n_1 and n_3 but is not a path between them (Non-example 2.38, Non-example 2.34). In the graph D there is no path between n_1 and n_3 at all.

5.1 The three conventions in one table

In plain words. The same two rules get different names in different books. One rule says never drive a road twice. The other says never pass a house twice. Our textbook, English books and our course do not always use the same word for the same rule. So here is one table that puts the words side by side.

The textbook's rows are its Definitions 1.2.10 to 1.2.15 [9]. The English words for walk, trail, path and cycle are Grinberg's Definitions 2.9.1(c), 3.4.5 and 2.10.1(b) [3]; "closed trail" is this paper's own name for the English counterpart of the textbook's cycle, since the sources checked give no single standard word for it. Wikipedia says that a path has distinct vertices "by most definitions" [8], and also records that some authors use "path" for walks with repeated vertices and "simple path" for those with distinct vertices [8]. That is the textbook's pattern, so the course convention is one of at least two in use.

In plain words. The word changes how many routes we count. It does not change whether a route exists. In the board village there is one route from the first house to the last. But there are three trips of at most four roads, and two of them turn back.

Remark 5.5 (The number of objects depends on the convention). Under the course convention, the number of paths from n_1 to n_3 in B is 1. The walks of length at most 4 from n_1 to n_3 are $1, 2, 3$ (length 2), $1, 2, 1, 2, 3$ and $1, 2, 3, 2, 3$ (length 4); only the first is a path or a trail (by script). So the number of walks, the number of trails and the number of paths are different numbers in general, and the entries of A^k give only the first (Proposition 3.10). The existence statement used by Definition 2.55 is the same under every convention (Proposition 2.59).

Table 7: The three naming conventions. "Course" is the assignment text's statement for the word path; the course convention for cycle is not stated, so this paper follows the textbook for cycle and writes "simple cycle" for the English "cycle". Sources: textbook [9], Grinberg [3], Wikipedia [8].

Property of the walk	Textbook word (number)	English word	Course and this paper
any list of steps along edges	walk (1.2.10)	walk	walk
walk with first vertex equal to last	closed walk (1.2.11)	closed walk	closed walk
no repeated edge; vertices may repeat	"path" (1.2.12)	trail	**trail**
no repeated vertex (so no repeated edge)	"simple path" (1.2.13)	path	**path**
closed, no repeated edge; vertices may repeat	"cycle" or "closed path" (1.2.14)	closed trail	cycle (textbook meaning)
closed, no repeated vertex except first and last, length at least 3	"simple cycle" (1.2.15)	cycle	simple cycle

6 Worked example: the board graph and a graph with one edge

This section writes out every matrix for two graphs and applies Theorem 3.5 and Theorem 4.6 to them. The two graphs are the board graph B (the matrix on the board, Figure 1, left) and the graph D with one edge (Figure 3). Every matrix and count below was computed by script, and the script outputs were checked against independent code before the numbers were written (Appendix A.7).

6.1 Powers of the board matrix

> **In plain words.** We take the village from the board. It has three houses in a row. The first and last houses each have one road. The middle house has two roads. We count trips of two roads, then of three roads. We list every trip, so the reader can see that each count is right.

Example 6.1 (The board matrix and its powers)**.** The graph B has vertices n_1, n_2, n_3 and edges n_1n_2 and n_2n_3. Its adjacency matrix and powers, copied from the script output, are

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad A^3 = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix},$$

$$A^4 = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 2 \end{pmatrix}, \qquad A^5 = \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 4 \\ 0 & 4 & 0 \end{pmatrix}.$$

The matrix A is symmetric with zero diagonal, as Proposition 2.13 requires; its row sums $1, 2, 1$ are the degrees, and the sum of all its entries is $4 = 2|E|$. Further, $A^3 = 2A$ and $A^{k+2} = 2A^k$ for $k = 1, \ldots, 5$ (by script), and the diagonal of A^2 is $(1, 2, 1)$, the degrees, with sum $4 = 2|E|$ (Corollary 3.9(d)). Each power is symmetric (Corollary 3.9(c)).

The board matrix is the matrix of a graph with two edges. The board drawing also shows an edge 1–3 and an arrowhead; they are not in the matrix (Table 3), and this section follows the matrix.

Every walk listed. For every entry of A, A^2 and A^3 the script lists the walks and checks that their number equals the entry (labels $1, 2, 3$ stand for n_1, n_2, n_3).

- Length 1: 1-2 and 2-3, and their reverses.
- Length 2: from 1 to 1: 1-2-1 (one walk). From 1 to 3: 1-2-3 (one). From 2 to 2: 2-1-2 and 2-3-2 (two). From 3 to 3: 3-2-3 (one). All other entries of A^2 are 0.
- Length 3: from 1 to 2: 1-2-1-2 and 1-2-3-2 (two). From 2 to 3: 2-1-2-3 and 2-3-2-3 (two). The entries $(2, 1)$ and $(3, 2)$ follow by reversing the walks.

The induction step of Theorem 3.5 can be checked on three entries, by the sum over the second-last vertex n_r (script output):

$$(A^2)_{22} = \sum_r A_{2r}A_{r2} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2,$$
$$(A^2)_{11} = \sum_r A_{1r}A_{r1} = 0 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 = 1,$$
$$(A^3)_{12} = \sum_r (A^2)_{1r}A_{r2} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2.$$

In the last line the term $r = 1$ is the walk 1-2-1 of length 2 extended by the step to n_2, giving 1-2-1-2, and the term $r = 3$ is the walk 1-2-3 extended by the step to n_2, giving 1-2-3-2. The walk 1-2-1, counted by $(A^2)_{11}$, is neither a path nor a trail: this is the counterexample of Proposition 3.10.

Walks, trails and paths by length. Table 8 counts all walks, trails, paths and closed walks of each length in B, over all pairs of start and end vertices, by enumeration. The totals in the walk column equal the sums of all entries of A^k: $4, 6, 8, 12$ for $k = 1, 2, 3, 4$ (script). From $k = 3$ on there is no trail (it would need three distinct edges, and B has two) and no path (it would need four distinct vertices, and B has three). The walk numbers keep growing; this is the warning of Proposition 3.10 in numbers.

Table 8: All walks, trails, paths and closed walks of length k in the board graph B, counted over all start and end vertices (by script). Length 0 counts the three single vertices.

k	walks	trails	paths	closed walks
0	3	3	3	3
1	4	4	4	0
2	6	2	2	4
3	8	0	0	0
4	12	0	0	8

6.2 The test on the board graph (connected)

In plain words. Take three houses in a row. A road joins the first and the second house. Another road joins the second and the third. The first and the third share no road. Can you still go from one to the other? Yes, through the middle house. We expect the check to say connected.

Example 6.2 (Matrix test on B, $N = 3$, $N - 1 = 2$)**.** Every product was computed by script. With $N - 1 = 2$ the sum is $S = A + A^2$, and the second form is $(I + A)^2$:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix},$$

$$S = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad (I + A)^2 = \begin{pmatrix} 2 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 2 \end{pmatrix}.$$

S has no zero entry off the diagonal (in fact none anywhere), so by Theorem 4.6 the graph is connected. $(I + A)^2$ has no zero entry, so by Proposition 4.16 the graph is connected. The two answers agree, and breadth-first search from vertex 1 reaches $\{1, 2, 3\}$ and agrees too. The entry A_{13} is 0, so no road joins n_1 and n_3, yet $(A^2)_{13} = 0 \cdot 0 + 1 \cdot 1 + 0 \cdot 0 = 1$: exactly one walk of length 2 joins them, namely $1, 2, 3$. The entry of $(I + A)^2$ at $(1, 3)$ is $1 = \binom{2}{2}(A^2)_{13}$, since in formula (3) both I and A have the entry 0 at $(1, 3)$. As a check, $(I + A)^2 = I + 2A + A^2$ holds entry by entry (script). *[derived here]*

By Corollary 4.10(b), the smallest k with $(A^k)_{ij} > 0$ is the distance: $d(1, 2) = d(2, 3) = 1$ from A, and $d(1, 3) = 2$ from A^2, in agreement with Example 2.50.

6.3 The test on the graph with one edge (disconnected)

> **In plain words.** Now take three houses again. Only one road exists, between the first and second house. The third house stands alone. Nobody can walk to it. We expect the check to say not connected. The check should also show that the third house is a piece by itself.

Example 6.3 (Matrix test on D, $N = 3$, $N - 1 = 2$)**.** The graph D has the single edge $n_1 n_2$. From the script,

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$S = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad (I + A)^2 = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The off-diagonal zeros of S are at $(1, 3)$, $(2, 3)$, $(3, 1)$ and $(3, 2)$, and the zeros of $(I + A)^2$ are at the same four places. So by Theorem 4.6 and Proposition 4.16 the graph is not connected. By Corollary 4.10(a), vertices 1 and 2 lie in one component and vertex 3 in another: the components are $\{1, 2\}$ and $\{3\}$, which agrees with breadth-first search. Higher powers do not help: the entries $(1, 3)$ and $(2, 3)$ of A^k are 0 for $k = 1, \ldots, 6$ (script); for instance

$$A^3 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = A.$$

In fact no walk reaches vertex 3 at all. *[derived here]*

The test on all small graphs. As a check of the statements, not as a proof, the script compared the sum test, the form with $I + A$ and the component statement of Corollary 4.10(a) with breadth-first search on every labelled simple graph with $N = 2, \ldots, 6$ vertices (Table 10). All agree.

Table 9: Outcome of the tests on the two graphs. Both forms of the matrix test and breadth-first search give the same answer.

Graph	Matrix test (Theorem 4.6)	Form with $I + A$ (Proposition 4.16)	Components
B, edges 12, 23	connected	connected	$\{1, 2, 3\}$
D, edge 12	not connected	not connected	$\{1, 2\}$, $\{3\}$

Table 10: Exhaustive check over all labelled simple graphs with N vertices (by script). "Sum test agrees" counts the graphs on which the test of Theorem 4.6 gave the same answer as breadth-first search. The last column counts the graphs on which the same test with only $N - 2$ powers gives a wrong answer (Proposition 4.17).

N	graphs	connected	sum test agrees	wrong with $N - 2$ powers
2	2	1	2	1
3	8	4	8	3
4	64	38	64	12
5	1,024	728	1,024	60
6	32,768	26,704	32,768	360

7 Discussion: limitations and failure cases

7.1 What the sources and photographs leave open

- **The mark above A.** The photographs do not show clearly whether the mark above A in Question 1 is a k, an n, or quotation marks (Section 1.3, item 2). The paper reads it as k because the assignment text does. The answer does not change with the letter, since Theorem 3.5 holds for every positive integer exponent *[interpretation, qualified]*. What stays open is only what the lecturer wrote.

- **The drawing and the matrix.** The paper follows the matrix and reports the extra edge 1–3 and the arrowhead (Table 3). If the lecturer meant the triangle T, the example would change but no theorem would: the statements hold for every simple undirected graph. The paper does not claim which graph was meant.

- **The meaning of "path".** That the course means the textbook's simple path is stated by the assignment text and is not on the board. If the course meant the textbook's Definition 1.2.12, then "path" is a trail, and the answer to Question 3 changes in words. The answer to Question 2 does not change, by Proposition 2.59. Counts of paths would change; counts of walks would not. *[qualified]*

- **What the lecturer expected for Question 2.** The board shows only the question, not the answer. The paper gives the sum test, the form with $I + A$, and breadth-first search, and does not say which one the lecturer intended.

- **The textbook beyond the pages checked.** The statement that the textbook contains neither the counting theorem nor a matrix test is *[qualified]*: it rests on the pages checked and on a full-text search (Section 1.6), not on a page-by-page reading of the whole book. It does not affect the proofs, which do not depend on it.

7.2 Limits of the results

- **Simple graphs.** Both proofs are for finite simple undirected graphs. Directed graphs and graphs with loops or repeated edges are outside the paper, and the connectivity test

should not be applied to them.

- **Only as good as two results.** The test rests on Theorem 3.5 and Lemma 2.52. The first is stated by several sources and proved here. The second is the textbook's Lemma 1.2.1 for $u \neq v$, which has no length bound; the bound comes from Grinberg's Corollary 2.9.7, and the paper proves the lemma with the bound itself (Section 2.3).
- **The test answers yes or no.** The sum S does not list a path between two vertices. Corollary 4.10 gives the components and the distances, and nothing more. Eigenvalue methods are outside the scope of this paper and are not discussed.
- **Finite checks are not proofs.** The brute-force comparisons (Appendix A.6) cover $N \leq 5$ for walk counting and $N \leq 6$ for the test. They support the proofs and do not replace them.
- **Source proofs.** Two sources state the connectivity test with incomplete proofs, and one states the counting theorem with "paths" in place of walks (Section 1.6, Section 3.4); that is why the paper proves both itself.

7.3 Failure cases of the test

Each of the following gives a wrong answer on a small graph.

- Stopping at A^{N-2}. On the row of four houses the $(1, 4)$ entry is 0 (Example 4.18). The graphs on which this fails are exactly the labelled rows of houses (Remark 4.19).
- Using the single power A^{N-1}, or A^2 alone for $N = 3$ (Non-example 4.4).
- Requiring positive diagonal entries of S. For $N = 2$ and one edge the diagonal is zero although the graph is connected (Remark 4.11).
- Using subtraction, or arithmetic modulo a number, in place of ordinary integer arithmetic. The step "a sum is positive exactly when a term is" needs non-negative integers (Theorem 4.6, *why the argument is valid*).
- Reading $(A^k)_{ij}$ as the number of paths (Proposition 3.10).

7.4 Cost, in brief

[estimate] Reaching A^{N-1} from A takes $N - 2$ matrix products, and the school method of multiplication uses about N^3 scalar multiplications for each product of two $N \times N$ matrices; faster methods were not examined. Breadth-first search with the graph stored as adjacency lists takes time at most a fixed multiple of $N + |E|$ [10], a single pass over the graph. The matrix test is therefore useful for its clean statement about the powers of A, not for speed *[interpretation]*.

8 Conclusion: the three answers

Question 1: what does $(A^k)_{ij}$ represent? For the adjacency matrix A of a finite simple undirected graph and an integer $k \geq 1$, the entry $(A^k)_{ij}$ is the number of *walks* of length k from n_i to n_j (Theorem 3.5). The proof is by induction on k, splitting the walks by their second-last vertex. The number counts walks, not paths: a walk may repeat vertices and edges, so $(A^2)_{ii} = \deg(n_i) > 0$ although no path of length 2 returns to n_i (Proposition 3.10). On the board graph, $(A^3)_{12} = 2$ while the only path from n_1 to n_2 has length 1. The reading of the mark above A as k is an interpretation (Section 1.3); the theorem holds for any positive integer exponent.

Question 2: how can one tell from A whether the graph is connected? For $N \geq 2$, the graph is connected if and only if every entry off the diagonal of $S = A + A^2 + \cdots + A^{N-1}$ is positive (Theorem 4.6). Equivalently, $(I + A)^{N-1}$ has no zero entry (Proposition 4.16). The proof uses walk counting, the lemma that every walk contains a path, and the bound that a path has at most $N - 1$ edges. The exponent $N - 1$ cannot be lowered to $N - 2$ (Proposition 4.17). On the board graph the test says connected, and on the graph with the single edge 12 it says not connected, with the components $\{1, 2\}$ and $\{3\}$ (Section 6). Breadth-first search gives the same answers (Section 4.6).

Question 3: what is a path between n_i and n_j? In the course convention, a path between n_i and n_j is a walk $w_0, \ldots, w_k$ with $w_0 = n_i$, $w_k = n_j$ and pairwise distinct vertices (Definition 5.1). It is the textbook's simple path, Definition 1.2.13, and the English path. The textbook's own word "path", Definition 1.2.12, means a walk with no repeated edge; this paper calls it a trail (Table 7). That the course uses this meaning comes from the assignment text and is *[qualified]*. Whether a path exists between two vertices does not depend on the choice (Proposition 2.59).

What the paper did not settle. The board's mark above A, the extra edge and arrowhead in the board drawing, and the lecturer's intended answer to Question 2 remain open (Section 7.1). The statement that the textbook contains no walk-counting theorem and no matrix test is *[qualified]*: it rests on the pages checked and a full-text search (Section 1.6).

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[4] *Lecture whiteboard photo 1.* Local file `sources/whiteboard-1.jpg`. Photograph of the lecture board, supplied by the user. n.d.

[5] *Lecture whiteboard photo 2.* Local file `sources/whiteboard-2.jpg`. Second photograph of the same board at a slightly different angle, supplied by the user. n.d.

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A Derivations, cross-checks and scripts

This appendix holds the longer derivations and the scripts that produced every number. The body of the paper gives the proofs of Theorems 3.5 and 4.6 and Lemma 2.52. The appendix adds the parts that the body only states: the reduction of walks to vertex lists, the general counterexample behind Proposition 3.10, the proof of Proposition 4.16, the distance results used in Remark 2.49 and Corollary 4.10, the chain of premises behind Theorem 4.6, and the finite checks.

A.1 Walks as vertex lists, and the induction step

In the textbook's sense a walk lists vertices and edges alternately, each edge joining the two vertices beside it (Definition 2.23). Take a list $(w_0, \dots, w_k)$ of vertices in which each consecutive pair is adjacent. For each t there is an edge $w_{t-1}w_t$, and it is the only edge with these two ends, because E is a set of unordered pairs (Definition 2.1). So the list determines the walk, by $e_t = w_{t-1}w_t$. Conversely, a walk determines its list of vertices. The two assignments are inverse to each other, so the walks of length k from n_i to n_j are in one to one correspondence with the lists $(w_0, \dots, w_k)$ with $w_0 = n_i$, $w_k = n_j$ and consecutive vertices adjacent. This is the reduction used in the proof of Theorem 3.5. *[derived here]*

The route of the proof, induction on k with the sum over the second-last vertex, is the one in the University of Galway notes and in Grinberg's proof of his Theorem 4.5.10 [2, 3]. Grinberg also treats $k = 0$, with $A^0 = I$ and one walk of length 0 from a vertex to itself [3]; the board question needs only $k \geq 1$.

The induction step, in words and in symbols, is the identity

$$W_k(i, j) = \sum_{r=1}^{N} W_{k-1}(i, r)\, A_{rj}, \tag{4}$$

where $W_k(i, j)$ is the number of walks of length k from n_i to n_j. The term for r counts the walks of length $k - 1$ from n_i to n_r that can be extended by the step to n_j; the factor A_{rj} is 1 when that step exists and 0 otherwise. The three hand checks on the board graph are written out in Section 6.1.

A.2 The walk-versus-path counterexample in general

Proposition 3.10(a) says: if n_i has a neighbour n_r, then the walk n_i, n_r, n_i is counted by $(A^2)_{ii}$, and it is neither a path nor a trail. The argument has three parts, each used in the proof in the body. First, the walk repeats n_i, so its vertices are not pairwise distinct. Second, it uses the edge $n_i n_r$ twice, so its edges are not pairwise distinct. Third, a path from a vertex to itself has pairwise distinct vertices, so it has length 0, not 2. Under each of the three conventions of Table 7 there is therefore no path or trail of length 2 from n_i to n_i, while $(A^2)_{ii} = \deg(n_i) \geq 1$.

On the board graph B the script also counts the closed walks from n_1 to n_1. For $k = 1, 2, 3, 4$ there are $0, 1, 0, 2$ of them, and none of these is a path or a trail. The 2 for $k = 4$ are 1-2-1-2-1 and 1-2-3-2-1, in agreement with $(A^4)_{11} = 2$ in Example 6.1. The walks from n_1 to n_2 are in Table 5. The identity $A^{k+2} = 2A^k$ follows from $A^3 = 2A$ (checked by script for $k = 1, \ldots, 5$) by multiplying with A^{k-1}.

A.3 The chain of premises behind the connectivity test

The proof of Theorem 4.6 is written in the body. Table 11 lists each step with the result it uses, so that the proof can be audited without reading it twice.

Table 11: The proof of Theorem 4.6, step by step.

Step	Statement	Relies on
1	G is connected iff every pair of vertices is joined by a path	Definition 2.55, Proposition 2.59
2	A path between different vertices has length between 1 and $N - 1$	Corollary 2.54
3	A walk of length k from n_i to n_j exists iff $(A^k)_{ij} > 0$	Theorem 3.5, Corollary 3.9(b)
4	A sum of non-negative integers is positive iff a term is positive	Corollary 3.9(a)
5	A walk from n_i to n_j contains a path from n_i to n_j	Lemma 2.52

A.4 The form with $I + A$ and the exponent m

We prove Proposition 4.16. Let $m \geq N - 1 \geq 1$.

Expansion. Write $(I + A)^m = (I + A)(I + A) \cdots (I + A)$ with m factors. Expanding the product gives one term for each choice of I or A from each factor, with the chosen matrices multiplied in order. Because $IX = XI = X$ for every matrix X, a term with k letters A and $m - k$ letters I equals A^k, wherever the letters stand. The number of choices with exactly k letters A is the binomial coefficient $\binom{m}{k}$. Therefore formula (3) holds:

$$(I + A)^m = \binom{m}{0} I + \binom{m}{1} A + \binom{m}{2} A^2 + \cdots + \binom{m}{m} A^m,$$

with $A^0 = I$. (This is the binomial theorem for the commuting matrices I and A.) For $m = 2$ it reads $(I + A)^2 = I + 2A + A^2$, which the script confirmed entry by entry on the board graph and on D.

Off-diagonal entries. Let $i \neq j$. The identity matrix has 0 at (i, j), so by (3)

$$\big((I + A)^m\big)_{ij} = \sum_{k=1}^{m} \binom{m}{k} (A^k)_{ij}.$$

Every coefficient $\binom{m}{k}$ with $1 \le k \le m$ is a positive integer, and every $(A^k)_{ij}$ is a non-negative integer (Corollary 3.9(a)). So the sum is positive if and only if at least one $(A^k)_{ij}$ with $1 \le k \le m$ is positive, that is, if and only if there is a walk of length k from n_i to n_j for some k with $1 \le k \le m$ (Corollary 3.9(b)).

If $S_{ij} > 0$, such a $k \le N-1 \le m$ exists by Proposition 4.5. Conversely, if a walk from n_i to n_j exists, Lemma 2.52 gives a path from n_i to n_j, of length k' with $1 \le k' \le N-1$ by Corollary 2.54; this path is a walk of length k', so $(A^{k'})_{ij} > 0$ and $S_{ij} > 0$. So the off-diagonal entry is positive exactly when $S_{ij} > 0$, which is item (a).

Diagonal entries. The term $\binom{m}{0} I$ contributes 1 to every diagonal entry and the other terms are non-negative, so every diagonal entry is at least 1, whatever the graph is. This is item (b).

Conclusion. By Theorem 4.6, G is connected if and only if $S_{ij} > 0$ for all $i \ne j$. By (a) and (b) this holds if and only if $(I + A)^m$ has no zero entry. *[derived here]* Bai gives the idea in item 3 of his proof sketch [1].

In the proof any exponent $m \ge N-1$ works, not only $m = N-1$; the script confirmed this on all labelled graphs with $N \le 6$ (Appendix A.6).

A.5 Distance: shortest walk, shortest trail, shortest path

> **In plain words.** Look at the shortest trip between two houses. It never passes a house twice. If it did, we could cut out the loop and get a shorter trip. That cannot be. So the shortest trip is also a route. It also never drives a road twice.

Proposition A.1. *For vertices u, v of a simple graph, the length of a shortest (u,v)-walk, of a shortest (u,v)-trail and of a shortest (u,v)-path are equal. Each is $+\infty$ when the corresponding set is empty, and the three sets are empty together. So the distance of Definition 2.48 is the same whether "path" is read as trail or as path, and it equals the length of a shortest walk.*

Proof. Every path is a trail (Proposition 2.39) and every trail is a walk. So the set of lengths of paths is contained in the set of lengths of trails, which is contained in the set of lengths of walks. Hence the minimum over walks is at most the minimum over trails, which is at most the minimum over paths. By Lemma 2.52, a shortest walk contains a path of at most the same length, so the minimum over paths is at most the minimum over walks. Therefore the three minima are equal. If there is no walk, there is no trail and no path. If there is a walk, Lemma 2.52 gives a path, and so the three sets are non-empty together. Grinberg makes the same remark [3, Section 2.9.4, p. 40]. □

> **In plain words.** The distance has three easy features. It is never negative, and it is zero only from a house to itself. It is the same in both directions. And a detour through a third house is never shorter than the shortest trip.

Proposition A.2. *The three properties of Remark 2.49 follow from Definition 2.48.*

Proof. By Proposition A.1 we may compute d by shortest walks. (1) Lengths are $0, 1, 2, \ldots$ or $+\infty$, so $d \ge 0$. The walk consisting of u alone has length 0, so $d(u,u) = 0$. If $d(u,v) = 0$ there is a walk of length 0 from u to v; it is the single vertex, so $v = u$. (2) If $u_0, \ldots, u_k$ is a (u,v)-walk, then $u_k, \ldots, u_0$ is a (v,u)-walk of the same length, since each step is an edge and edges are unordered. So the sets of walk lengths from u to v and from v to u are equal, and so are their minima. (3) If $d(u,w)$ or $d(w,v)$ is $+\infty$, the right side is $+\infty$ and the inequality holds. Otherwise join a shortest (u,w)-walk and a shortest (w,v)-walk at w. The result is a (u,v)-walk of length $d(u,w) + d(w,v)$, and the shortest (u,v)-walk is no longer than this. □

A.6 Cross-checks by script

These are tests of the statements by computer. They support the proofs and do not replace them. All were run by the scripts of Appendix A.7.

- **Walk counting**. For every labelled simple graph on $N = 2, 3, 4, 5$ vertices (2, 8, 64, 1024 graphs), the script enumerated all walks by depth-first search and compared their number with the entries of A^k (k up to 5 for $N \leq 4$, up to 3 for $N = 5$): 40, 360, 5120 and 76800 cases, all equal.
- **Walk-to-path lemma and its consequences**. The construction of Lemma 2.52 produced a path with the same ends, no longer than the walk, with vertices taken from the walk, for all 73 walks of length 0 to 6 in B, all 381 in T and all 4695 in G_{tb}: no failure. On 300 random graphs with N from 2 to 7 (seed 20261027) and 192696 walks of length 0 to 5 there was no failure of the lemma, no path with a repeated edge (Proposition 2.39), no path longer than $N - 1$ (Corollary 2.54), and no disagreement between breadth-first distance and the shortest walk, trail and path (Proposition A.1).
- **The connectivity test**. On all labelled simple graphs with $N = 2, \ldots, 6$ vertices (Table 10), the sum test, the form with $I + A$, and the component statement of Corollary 4.10(a) all agreed with breadth-first search. The same table counts the graphs on which the test with $N - 2$ powers fails: 1, 3, 12, 60, 360 for $N = 2, \ldots, 6$ (for $N = 2$ the sum with 0 powers is empty).

A.7 Scripts and how to reproduce

Every number in the paper came from one of the scripts in `appendices/scripts/`. They use pure Python integers, so there is no rounding, and no library. They are copies of the module scripts written during the research, with their saved outputs; the paper's own script was added. Run each one from the paper folder with `python3 -I`, which keeps the interpreter from loading files from the current directory.

Table 12: The scripts and what each one computes. The saved output of each module script has the same name with `_output.txt` or `.out.txt` in place of `.py`.

Script	Computes
`m1_walks.py`	matrix powers of the board matrix, explicit walk lists, the sums of the induction step, walks against paths and trails, the brute-force check of walk counting
`m2_walks.py`	classification of the example sequences, counts of walks, trails, paths and closed walks, the trace of the walk-to-path lemma, exhaustive and random tests, distances
`m3_connectivity.py`	the sum test and the $(I + A)^{N-1}$ test on the board graph, on D and on the row of four houses, the exhaustive check against breadth-first search, and operation counts (about 35 seconds)
`generate_paper_numbers.py`	reads the three saved outputs, recomputes every matrix and table with independent code, asserts equality, and writes `appendices/generated_numbers.tex`, which defines the matrices and table bodies used in the text

For example, from the paper folder:

```
python3 -I appendices/scripts/generate_paper_numbers.py
```

The matrices in Section 6, Example 4.18 and the tables of Tables 5, 6, 8 and 10 are macros from that file, so none was typed by hand.

B Course terms and textbook numbers

This appendix lists every term of the course that the paper uses, with the number and printed page in the course textbook [9] and the English word. Page numbers are the printed page numbers of the textbook; the PDF page is one higher. The textbook's own word is given in Latin letters as "path", "simple path" and so on; its Armenian words are not reproduced. Terms that the pages checked give no number for are marked "none".

Table 13: Course terms, textbook numbers and pages, English words, and where they are defined in this paper.

Term in this paper	Textbook number	Printed page	Textbook word if different	Defined in
simple undirected graph	Definition 1.1.1	9	graph	Definition 2.1
adjacent	Definition 1.1.4	11	—	Definition 2.5
adjacency matrix	none (after Def. 1.1.6)	11	$A(G)$	Definition 2.9
degree	Definition 1.2.1	16	$d_G(v)$, $d(v)$	Definition 2.16
sum of degrees	Theorem 1.2.1 (Euler)	16	—	Proposition 2.20
walk	Definition 1.2.10	21	walk	Definition 2.23
closed walk	Definition 1.2.11	22	closed walk	Definition 2.27
trail	Definition 1.2.12	22	"path"	Definition 2.31
path	Definition 1.2.13	22	"simple path"	Definition 2.35
cycle	Definition 1.2.14	22	"cycle" or "closed path"	Definition 2.40
simple cycle	Definition 1.2.15	22	"simple cycle"	Definition 2.44
distance	Definition 1.2.16	22–23	distance	Definition 2.48
walk contains a path	Lemma 1.2.1 (for $u \neq v$)	23	—	Lemma 2.52
connected	Definition 2.1.1	48	connected	Definition 2.55
connected component	none (prose)	49	connectivity component	Definition 2.60
connected iff one component	Remark 2.1.1	49	—	Remark 2.61
path between n_i and n_j	Definition 1.2.13	22	"simple path"	Definition 5.1

Figures of the textbook used. Fig. 1.2.5 (printed p. 22) is the source of the examples for walks, trails, paths and cycles; Figure 2 draws the seven edges that those examples force. Fig. 2.1.1 (p. 48) shows three connected graphs and Fig. 2.1.2 (p. 49) a graph with four components [9].

Results that the textbook does not contain. The walk-counting theorem (Theorem 3.5) and the matrix test for connectivity (Theorem 4.6) are not in the pages checked and were not found by a full-text search of the book (Section 1.6), so their absence from the book is *[qualified]*. For the walk-counting theorem the paper cites the University of Galway notes, and for the statement of the test Paramadevan and Sotheeswaran and Weisstein; the proofs are the paper's own.

The board's letters. The board writes (G, V) with V the edge set, $e_{ij} \in V$, $|G| = N$ and n_i [4, 5]; the textbook writes $G = (V, E)$, $e \in E$, n and v_i. The paper uses $G = (V, E)$, N and n_i (Table 4).